

ST. JOSEPH'S COLLEGE OF COMMERCE

COST MANAGEMENT – I



Fully Worked Solutions with Working Notes & Method Steps

Course B.Com Regular — Semester V
Modules 2 · Marginal & Absorption Costing
3 · Cost-Volume-Profit Analysis
4 · Relevant Costing
5 · Budget & Budgetary Control
Source College workbook problems + classmate's
handwritten solutions (all 28 pages)
Problems 55 solved

Every figure below is arithmetically re-verified. Where the handwritten notes contain a slip, the corrected figure is used and the slip is flagged in a red box.

HOW TO SCORE FULL MARKS — presentation rules that examiners look for

Indian B.Com cost accounting papers award marks for **presentation and workings**, not just the final figure. In most university schemes a problem carries roughly 60–70% for the correct format & workings and only 30–40% for the final answer. That means **a wrong final figure with perfect format still scores well**, and a right figure with no workings loses marks. Follow these 14 rules on every single problem.

A. The non-negotiables (do these before you compute anything)

- **1. Write a full title on every statement.** Never just "Solution". Write e.g. **Statement of Profit under Marginal Costing for the quarter ended 31st March**. A statement without a heading loses the format mark.
- **2. Rule the columns first.** Draw **Particulars** on the left, then amount columns. Put **Rs.** at the head of every money column and **Units** at the head of every quantity column. Never mix units and rupees in one column.
- **3. Use the prescribed proforma.** Marginal costing = vertical statement (Sales – VC = Contribution – FC = Profit). Absorption costing = Sales – COGS = Gross Profit – Other fixed = Net Profit. Ledger items (Process A/c, Abnormal Loss A/c) = **T-format with To / By**, never vertical.
- **4. Deductions in brackets** and write the word **Less: .** Additions get **Add: .** This is what makes the examiner able to follow you.

B. Working notes — where most of the marks actually sit

- **5. Number your working notes** W.N.1, W.N.2, W.N.3 and place them *after* the main statement (or before, but consistently).
- **6. Cross-reference every derived figure.** If a figure in the main statement is not given in the question, it must carry a tag like **(W.N.2)**. Rule of thumb: **an examiner should never have to ask "where did this number come from?"**
- **7. Show formula → substitution → answer, on three lines.** Example:
$$\begin{aligned} \text{P/V Ratio} &= \text{Contribution} / \text{Sales} \times 100 \\ &= 50,000 / 2,00,000 \times 100 \\ &= 25\% \end{aligned}$$

Writing only "**= 25%**" forfeits the formula mark even when correct.
- **8. Show the per-unit column.** In flexible budgets, marginal costing and absorption costing, always give **both** Per Unit and Total columns. It costs one extra column and earns marks.

C. Finishing the answer

- **9. Totals must tie.** Cast (add) both sides of every account and underline the total with a **double rule**. In a Process A/c the units column must balance as well as the amount column.
- **10. State assumptions in one line.** e.g. "Assumed that fixed overheads remain unchanged at the increased level of output" or "Selling & distribution expenses assumed to vary with sales value." An examiner cannot penalise a stated assumption.
- **11. Every decision question needs a written recommendation.** Compute, then write a sentence: Since the net relevant cost is lowest under 'Buy & hire out' (Rs.1,69,000), that alternative is recommended. The sentence itself carries 1–2 marks and students routinely skip it.
- **12. Give the reason with the decision** — "because contribution per limiting factor is highest", "because it is a sunk cost and hence irrelevant". Bare answers get partial credit only.
- **13. Round consistently** and say so once: "figures rounded to the nearest rupee". Do not round mid-way through a chain of calculations — round only at the end.
- **14. Underline final answers** (single rule) and box or double-underline the ultimate figure. Keep pencil for ruling tables, pen for figures.

Five habits that lose marks silently: (i) mixing fixed cost into contribution in a marginal costing statement; (ii) forgetting that **fixed cost per unit changes** with volume while **variable cost per unit stays constant**; (iii) valuing closing stock at total cost under marginal costing (it must be at **variable cost only**); (iv) omitting the units column in a process or budget account; (v) not writing the decision sentence in make-or-buy / accept-or-reject / shut-down problems.

MODULE 2 — Marginal Costing & Absorption Costing

Formula sheet (learn this before attempting any problem)

Marginal Cost Equation	Sales – Variable Cost = Contribution ; Contribution – Fixed Cost = Profit / Loss
Contribution	= Sales – Variable Cost or = Fixed Cost + Profit
P/V Ratio (5 ways — pick whichever the data allows)	(a) $(\text{Contribution} \div \text{Sales}) \times 100$ (b) $(\text{Contribution p.u.} \div \text{Sales p.u.}) \times 100$ (c) $(\text{Change in Contribution} \div \text{Change in Sales}) \times 100$ (d) $(\text{Change in Profit} \div \text{Change in Sales}) \times 100$ ← use when 2 years' sales & profit given (e) $100\% - \text{Variable Cost Ratio}$, where $\text{VCR} = (\text{VC} \div \text{Sales}) \times 100$
Break-Even Point	$\text{BEP (Rs.)} = \text{Fixed Cost} \div \text{P/V Ratio}$ or $(\text{FC} \div \text{Contribution}) \times \text{Sales}$ $\text{BEP (units)} = \text{Fixed Cost} \div \text{Contribution per unit}$
Margin of Safety	$\text{MOS (Rs.)} = \text{Actual Sales} - \text{BEP Sales}$ or $\text{Profit} \div \text{P/V Ratio}$ $\text{MOS (units)} = \text{Profit} \div \text{Contribution per unit}$ $\text{MOS \%} = (\text{MOS} \div \text{Actual Sales}) \times 100$
Sales for a target profit	$\text{Required Sales} = (\text{Fixed Cost} + \text{Desired Profit}) \div \text{P/V Ratio}$ $\text{Required Units} = (\text{Fixed Cost} + \text{Desired Profit}) \div \text{Contribution per unit}$
Target profit after tax	First gross it up: $\text{PBT} = \text{PAT} \div (1 - \text{tax rate})$. Then use the formula above with PBT.
Absorption rate	= Variable production cost p.u. + Fixed production cost p.u. where $\text{Fixed production cost p.u.} = \text{Budgeted fixed production OH} \div \text{Normal / budgeted output}$
Under / over absorption	$\text{Absorbed OH} = \text{Actual production} \times \text{Fixed OH rate p.u.}$ Absorbed < Actual OH → under -absorbed (deduct from profit); Absorbed > Actual OH → over -absorbed (add to profit)

The single most important rule in this module — stock movement decides which profit is higher

- **Closing Stock > Opening Stock** (inventory rising) → **Absorption profit is HIGHER** than marginal profit, because fixed overhead is carried forward inside closing stock instead of being expensed.
- **Closing Stock < Opening Stock** (inventory falling) → **Absorption profit is LOWER** than marginal profit, because fixed overhead sitting in opening stock is released into this period's cost.
- **No change in inventory → both profits are equal.**
- Reconciliation (always ask for full marks):

$$\text{Profit as per Marginal Costing} + \text{Fixed OH in Closing Stock} - \text{Fixed OH in Opening Stock} = \text{Profit as per Absorption Costing}$$
- Also remember: under **marginal** costing stock is valued at **variable cost only**; under **absorption** costing at **variable + fixed production cost**. Never at total cost including selling overheads.

Effect of a change in one variable (memorise this grid — it answers Q11-type questions instantly)

- Change in **Fixed Cost** → P/V ratio **unaffected**; BEP moves in the **same** direction as FC; MOS moves **inversely**.
- Change in **Variable Cost** → P/V ratio moves **inversely**; BEP moves **inversely** to P/V ratio.
- Change in **Selling Price** → affects **both** Sales and P/V ratio; SP up → P/V up → BEP down.

Problems

Q2 Find profit using the marginal cost technique

Given: Fixed Cost Rs.5,00,000 | Variable Cost Rs.10 p.u. | Selling Price Rs.15 p.u. | Output 1,50,000 units.

Method — where to start:

1. Multiply the per-unit figures by output to get totals. Do **not** multiply fixed cost by output.
2. Lay out the vertical marginal cost statement with a Per Unit column beside the Total column.
3. Contribution first, fixed cost only at the last step.

Statement of Profit under Marginal Costing

Particulars	Total (Rs.)	Per unit (Rs.)
Sales (1,50,000 × 15)	22,50,000	15
Less: Variable Cost (1,50,000 × 10)	(15,00,000)	(10)
Contribution	7,50,000	5
Less: Fixed Cost	(5,00,000)	-
Profit	2,50,000	-

Answer: Profit = **Rs.2,50,000**

Q3 Find the profit from Sales, Marginal Cost and BEP

Given: Sales Rs.80,000 | Marginal (variable) Cost Rs.60,000 | BEP Rs.60,000.

Method — the trick here is that fixed cost is hidden:

1. Contribution = Sales – Marginal Cost.
2. Use the BEP formula **backwards** to extract Fixed Cost, since BEP is given.
3. Profit = Contribution – Fixed Cost.

W.N.1 — Contribution

Contribution = Sales – Marginal Cost = 80,000 – 60,000 = **Rs.20,000** ($\therefore$ P/V ratio = 20,000/80,000 = 25%)

W.N.2 — Fixed Cost from the given BEP

BEP = (Fixed Cost ÷ Contribution) × Sales

60,000 = (FC ÷ 20,000) × 80,000

60,000 = FC × 4 $\Rightarrow$ **FC = Rs.15,000**

Answer: Profit = Contribution – Fixed Cost = 20,000 – 15,000 = **Rs.5,000**

Q4 Calculate the Break-Even Point

Given: Variable cost p.u. Rs.12 | Fixed Expenses Rs.60,000 | Selling Price p.u. Rs.18.

Method:

1. Contribution p.u. = SP – VC.
2. BEP (units) = FC ÷ Contribution p.u.
3. Convert to value by multiplying BEP units by SP — **always give both**, examiners expect it.

Contribution per unit = 18 – 12 = **Rs.6**

BEP (units) = Fixed Cost ÷ Contribution p.u. = 60,000 ÷ 6 = **10,000 units**

BEP (value) = 10,000 × 18 = **Rs.1,80,000**

Answer: BEP = **10,000 units** or **Rs.1,80,000**

Q5 Find the selling price so that BEP falls to 9,000 units

Given: Variable cost p.u. Rs.75 | Fixed Expenses Rs.2,70,000 | present SP Rs.100. Required: SP at which BEP = 9,000 units.

Method — work the BEP formula in reverse:

1. BEP is **given** this time, so first find the contribution per unit that produces it: Contribution p.u. = FC ÷ BEP units.
2. Then SP = Variable cost p.u. + required Contribution p.u.
3. Do **not** start from the present SP of Rs.100 — that is only background information.

Let the required selling price be x.

$$\text{BEP (units)} = \text{FC} \div (\text{x} - \text{VC p.u.}) \Rightarrow 9,000 = 2,70,000 \div (\text{x} - 75)$$

$$\text{x} - 75 = 2,70,000 \div 9,000 = 30$$

$$\text{x} = 30 + 75 = \text{Rs.105}$$

Answer: Selling price must be raised to **Rs.105 per unit** (required contribution Rs.30 p.u.).

Watch out: at the present price of Rs.100 the contribution is only Rs.25, giving BEP = 2,70,000/25 = 10,800 units. Do not stop at 10,800 — that is the *existing* BEP, not the answer asked for.

Q6 BEP in units, and the new BEP if selling price is cut by 10%

Given: Fixed — Depreciation Rs.1,00,000 + Salaries Rs.1,00,000 = Rs.2,00,000. Variable — Materials Rs.3 p.u. + Labour Rs.2 p.u. = Rs.5 p.u. Selling Price Rs.10 p.u.

Method:

1. Total the fixed items and total the variable items separately — show this grouping, it earns a mark.
2. Compute BEP at the existing price.
3. Recompute contribution at the reduced price, then the new BEP. Fixed cost does **not** change.

Break-even computation, before and after the price cut

Particulars	Existing	SP reduced by 10%
Selling price per unit	10	9
Less: Variable cost per unit (3 + 2)	(5)	(5)
Contribution per unit	5	4
Fixed Cost (1,00,000 + 1,00,000)	2,00,000	2,00,000
BEP (units) = FC ÷ Contribution p.u.	40,000	50,000

Answer: Existing BEP = **40,000 units**; after a 10% price cut BEP rises to **50,000 units**.

Reason to write down: a lower selling price reduces contribution per unit, so more units must be sold to recover the same fixed cost.

Q7 P/V ratio from two periods' sales and profit

Given: Period I — Sales Rs.20 lakhs, Profit Rs.2 lakhs. Period II — Sales Rs.30 lakhs, Profit Rs.4 lakhs.

Method — recognise the pattern:

1. When **two periods'** sales and profits are given and nothing else, you **must** use P/V Ratio = Change in Profit ÷ Change in Sales. This works because fixed cost is the same in both periods and therefore cancels out.
2. Once the P/V ratio is known, back out contribution, then fixed cost, then BEP.

P/V Ratio = (Change in Profit ÷ Change in Sales) × 100 = (2,00,000 ÷ 10,00,000) × 100 = **20%**

Useful extensions (commonly asked in the same question)

Step	Working	Result
Contribution, Period I	20% × 20,00,000	4,00,000
Fixed Cost	Contribution – Profit = 4,00,000 – 2,00,000	2,00,000
BEP (Rs.)	FC ÷ P/V ratio = 2,00,000 ÷ 0.20	10,00,000

Answer: P/V Ratio = **20%**; Fixed Cost = Rs.2,00,000; BEP = **Rs.10,00,000**

Q8 Calculate the break-even point from two years' data

Given: Year 1 — Sales Rs.80,000, Profit Rs.10,000. Year 2 — Sales Rs.90,000, Profit Rs.14,000.

Method:

1. P/V ratio from the change in profit over the change in sales.
2. Contribution of each year = P/V ratio × that year's sales.
3. Fixed Cost = Contribution – Profit. Verify it is the **same** in both years — this is your self-check.
4. BEP = FC ÷ P/V ratio.

P/V Ratio = (14,000 – 10,000) ÷ (90,000 – 80,000) × 100 = 4,000/10,000 × 100 = **40%**

W.N.1 — Contribution and Fixed Cost of each year

Particulars	Year 1 (Rs.)	Year 2 (Rs.)
Sales	80,000	90,000
Contribution (40% of sales)	32,000	36,000
Less: Fixed Cost	(22,000)	(22,000)
Profit (agrees with the question — check passed)	10,000	14,000

Answer: BEP = FC ÷ P/V ratio = 22,000 ÷ 0.40 = **Rs.55,000**

Self-check you must not skip: fixed cost derived from Year 1 and from Year 2 has to be identical (Rs.22,000 here). If the two differ, your P/V ratio is wrong — go back before continuing.

Q9 Find the Margin of Safety

Given: Profit Rs.30,000 | Marginal cost Rs.8 p.u. | Selling price Rs.10 p.u.

Method:

1. Contribution p.u. = SP – MC = 10 – 8 = Rs.2.
2. MOS in **units** = Profit ÷ Contribution p.u.
3. MOS in **value** = MOS units × SP (or directly Profit ÷ P/V ratio).

Contribution per unit = 10 – 8 = Rs.2 → P/V ratio = 2/10 = 20%

MOS (units) = Profit ÷ Contribution p.u. = 30,000 ÷ 2 = **15,000 units**

MOS (value) = 15,000 × 10 = **Rs.1,50,000** cross-check: Profit ÷ P/V ratio = 30,000 ÷ 0.20 = 1,50,000 ✓

Answer: Margin of Safety = **Rs.1,50,000** (15,000 units)

Q10 Find P/V Ratio, BEP and Margin of Safety

Given: Sales Rs.1,00,000 | Total cost Rs.80,000 | Fixed Cost Rs.20,000 | Net profit Rs.20,000.

Method — first split total cost:

1. Variable Cost = Total Cost – Fixed Cost. This is the step students forget.
2. Build the marginal cost statement, then read off the three answers.

Marginal Cost Statement

Particulars	Rs.
Sales	1,00,000
Less: Variable Cost (80,000 – 20,000)	(60,000)
Contribution	40,000
Less: Fixed Cost	(20,000)
Profit	20,000

Required	Formula & substitution	Answer
(a) P/V Ratio	$(\text{Contribution} \div \text{Sales}) \times 100 = (40,000 \div 1,00,000) \times 100$	40%
(b) BEP	$\text{FC} \div \text{P/V Ratio} = 20,000 \div 0.40$	Rs.50,000
(c) Margin of Safety	$\text{Actual Sales} - \text{BEP Sales} = 1,00,000 - 50,000$	Rs.50,000

Q11 Effect on P/V Ratio, BEP and MOS under seven different changes

Given (base position): Sales Rs.20,000 | Variable Cost Rs.10,000 | Fixed Cost Rs.6,000.

Method — how to attack a columnar "what-if" question:

1. Put the **base case in the first column** and each scenario in its own column. One big table, never seven separate statements.
2. Change **only** the line the scenario mentions; copy every other line unchanged. This is where marks are lost — e.g. in "10% decrease in VC" the Sales figure stays at 20,000.
3. Compute Contribution → Profit for every column, then add three summary rows: P/V Ratio, BEP, MOS.
4. For BEP always use **BEP = (FC ÷ Contribution) × Sales** rather than dividing by a rounded percentage — it avoids rounding error.

Comparative Statement of Profit under each situation (Rs.)

Particulars	Base	(i) FC –20%	(ii) FC +10%	(iii) VC –10%	(iv) VC +10%	(v) SP+10% FC+1,200	(vi) SP –10%	(vii) SP–10% & VC–10%
Sales	20,000	20,000	20,000	20,000	20,000	22,000	18,000	18,000
Less: Variable Cost	(10,000)	(10,000)	(10,000)	(9,000)	(11,000)	(10,000)	(10,000)	(9,000)
Contribution	10,000	10,000	10,000	11,000	9,000	12,000	8,000	9,000
Less: Fixed Cost	(6,000)	(4,800)	(6,600)	(6,000)	(6,000)	(7,200)	(6,000)	(6,000)
Profit	4,000	5,200	3,400	5,000	3,000	4,800	2,000	3,000
P/V Ratio	50%	50%	50%	55%	45%	54.55%	44.44%	50%
BEP (Rs.)	12,000	9,600	13,200	10,909	13,333	13,200	13,500	12,000
MOS (Rs.)	8,000	10,400	6,800	9,091	6,667	8,800	4,500	6,000

Conclusions to write below the table: (1) A change in fixed cost leaves the **P/V ratio unchanged**; it moves BEP in the same direction and MOS in the opposite direction. (2) A fall in variable cost **raises** the P/V ratio and lowers BEP — the most effective of all the changes here (profit Rs.5,000 from case iii). (3) Case (i), a 20% cut in fixed cost, gives the highest profit of **Rs.5,200** and the widest margin of safety of Rs.10,400. (4) Case (vi), a 10% price cut with no other change, is the worst outcome — profit falls to Rs.2,000 and MOS collapses to Rs. 4,500.

Correction to note (the handwritten pages round too early here): in case (v) the P/V ratio is $12,000/22,000 = 54.55\%$, so $\text{BEP} = (7,200/12,000) \times 22,000 = \text{Rs.13,200}$ and $\text{MOS} = \text{Rs.8,800}$ — not $13,333 / 8,667$. In case (vi) the ratio is $8,000/18,000 = 44.44\%$, so $\text{BEP} = (6,000/8,000) \times 18,000 = \text{Rs.13,500}$ and $\text{MOS} = \text{Rs.4,500}$ — not $13,636 / 4,364$. Using $\text{BEP} = (\text{FC} \div \text{Contribution}) \times \text{Sales}$ avoids this entirely.

Q12 P/V Ratio, BEP, profit at a higher sales level, and sales for a target profit

Given: Variable Cost Rs.6,00,000 | Fixed Cost Rs.3,00,000 | Net Profit Rs.1,00,000 | Sales Rs.10,00,000.

Method:

- Contribution = Sales – VC. Confirm that Contribution – FC equals the given profit — a free accuracy check.
- Parts (c) and (d) both rest on one idea: **variable cost stays a constant percentage of sales** (60% here), so the P/V ratio of 40% holds at every sales level.
- For (c) scale up the variable cost; for (d) use Required Sales = (FC + Desired Profit) ÷ P/V Ratio.

(a) P/V Ratio

Contribution = 10,00,000 – 6,00,000 = Rs.4,00,000

P/V Ratio = (4,00,000 ÷ 10,00,000) × 100 = **40%** check: 4,00,000 – 3,00,000 = 1,00,000 = given profit ✓

(b) Break-Even Point

BEP = FC ÷ P/V Ratio = 3,00,000 ÷ 0.40 = **Rs.7,50,000**

(c) Profit when sales are Rs.12,00,000

Particulars	Rs.	Working
Sales	12,00,000	given
Less: Variable Cost	(7,20,000)	60% of sales (sales rose 20%, so VC rises 20%)
Contribution	4,80,000	40% of 12,00,000
Less: Fixed Cost	(3,00,000)	unchanged
Profit	1,80,000	

(d) Sales required to earn a profit of Rs.2,00,000

Required Sales = (Fixed Cost + Desired Profit) ÷ P/V Ratio = (3,00,000 + 2,00,000) ÷ 0.40 = **Rs.12,50,000**

Proof: Sales 12,50,000 – VC 7,50,000 (60%) = Contribution 5,00,000 – FC 3,00,000 = Profit 2,00,000 ✓

Answers: (a) 40% (b) Rs.7,50,000 (c) Rs.1,80,000 (d) Rs.12,50,000

Q13 Seven-part question including a profit target after tax

Given (year 2025): Sales Rs.2,00,000 | Variable Cost Rs.1,50,000 | Contribution Rs.50,000 | Fixed Cost Rs.15,000 | Profit Rs.35,000.

Method — do them in this order, each part feeds the next:

- P/V ratio first (everything else depends on it). Note VC ratio = 75%, so P/V = 25%.
- BEP next.
- Parts (iv) and (v) use the **new** sales level of Rs.3,00,000 but the **same** P/V ratio and fixed cost.
- Part (vii) is the only tricky one: **gross up the after-tax profit to a before-tax profit first**, then use the target-profit formula.

Answers with full workings

Part	Formula & substitution	Answer
(i) P/V Ratio	(Contribution ÷ Sales) × 100 = (50,000 ÷ 2,00,000) × 100	25%
(ii) Break-Even Point	FC ÷ P/V Ratio = 15,000 ÷ 0.25 or (FC ÷ Contribution) × Sales = (15,000/50,000) × 2,00,000	Rs.60,000
(iii) Sales for a 40% P/V Ratio	Contribution ÷ desired P/V Ratio = 50,000 ÷ 0.40 i.e. the sales value at which the same contribution of Rs.50,000 would represent 40%	Rs.1,25,000
(iv) MOS if actual sales are Rs. 3,00,000	Actual Sales – BEP Sales = 3,00,000 – 60,000	Rs.2,40,000

Part	Formula & substitution	Answer
(v) Net Profit if sales are Rs. 3,00,000	Contribution = 25% × 3,00,000 = 75,000; less FC 15,000	Rs.60,000
(vi) Sales to earn a profit of Rs. 70,000	(FC + Desired Profit) ÷ P/V Ratio = (15,000 + 70,000) ÷ 0.25	Rs.3,40,000
(vii) Sales to earn Rs.70,000 <i>after</i> tax @ 60%	Step 1 — gross up: PBT = 70,000 ÷ (1 – 0.60) = Rs.1,75,000 Step 2 — (FC + PBT) ÷ P/V Ratio = (15,000 + 1,75,000) ÷ 0.25	Rs.7,60,000

Proof of part (vii) — write this out, it earns the verification mark

Sales	7,60,000
Less: Variable Cost (75% of sales)	(5,70,000)
<i>Contribution (25%)</i>	<i>1,90,000</i>
Less: Fixed Cost	(15,000)
Profit before tax	1,75,000
Less: Tax @ 60%	(1,05,000)
Profit after tax	70,000

The classic error in part (vii): putting Rs.70,000 straight into the target-profit formula. That gives Rs.3,40,000, which is the answer to part (vi), not (vii). Tax is charged **on profit**, so the profit figure used in the formula must always be the **pre-tax** figure.

Q14 Merger of two companies — break-even sales, capacity, profitability and a price increase

Given: Company I at 90% capacity — Sales 540, VC 396, FC 80. Company II at 60% capacity — Sales 300, VC 225, FC 50. (Rs. in lakhs)

Method — the compulsory first step everyone misses:

1. The two companies are at **different capacity levels**, so you cannot simply add them. **Restate both at 100% capacity first.** Sales and variable cost are scaled up; **fixed cost is not scaled** because it does not vary with activity.
2. Company I: $540 \times 100/90 = 600$ and $396 \times 100/90 = 440$. Company II: $300 \times 100/60 = 500$ and $225 \times 100/60 = 375$.
3. Add the two 100% columns to get the merged plant. Now compute the merged P/V ratio.
4. Everything else follows from that single P/V ratio.

W.N.1 — Restatement at 100% capacity and the merged position (Rs. in lakhs)

Particulars	Company I	Company II	Merged plant
Capacity utilisation	100%	100%	100%
Sales	600	500	1,100
Less: Variable cost	(440)	(375)	(815)
<i>Contribution</i>	<i>160</i>	<i>125</i>	<i>285</i>
Less: Fixed cost	(80)	(50)	(130)
Profit	80	75	155

P/V Ratio of the merged plant = $(285 \div 1,100) \times 100 = 25.91\%$

(a) Break-even sales of the merged plant and the capacity at that point

Break-even Sales = $FC \div P/V \text{ Ratio} = (130 \div 285) \times 1,100 = \text{Rs.501.75 lakhs}$

Break-even capacity = $(501.75 \div 1,100) \times 100 = 45.61\%$

(b) Profitability of the merged plant at 80% capacity

Particulars	Rs. lakhs	Working
Sales	880	80% of 1,100
Less: Variable cost	(652)	80% of 815
Contribution	228	
Less: Fixed cost	(130)	unchanged at all capacities
Profit	98	

Profitability = $(98 \div 880) \times 100 = 11.14\%$ of sales

(c) Sales turnover needed to earn a profit of Rs.75 lakhs

Required Sales = $(FC + \text{Desired Profit}) \div P/V \text{ Ratio} = (130 + 75) \div 0.2591 = (205 \div 285) \times 1,100 = \text{Rs.791.23 lakhs}$

Check: Contribution 25.91% of 791.23 = 205 – FC 130 = Profit 75 ✓ (capacity = $791.23/1,100 = 71.93\%$)

(d) Price increase needed to absorb a 5% rise in fixed overheads

1. Extra fixed cost = $5\% \times 130 = \text{Rs.6.50 lakhs}$.
2. To hold profit at Rs.75 lakhs, contribution must rise by the same Rs.6.50 lakhs.
3. A price rise adds to sales **without touching variable cost**, so the extra sales revenue needed is exactly Rs. 6.50 lakhs.
4. Express that as a percentage of the sales level in part (c).

% increase in selling price = $(6.50 \div 791.23) \times 100 = 0.82\%$

Answers: (a) BE sales Rs.501.75 lakhs at 45.61% capacity (b) Profit Rs.98 lakhs, profitability 11.14% (c) Rs. 791.23 lakhs (d) selling price must rise by **0.82%**

Q16 Profit under Absorption costing vs Marginal costing, two periods

Given: Variable production cost Rs.4 p.u.; Selling price Rs.6 p.u.; Fixed costs Rs.2,000 per period, of which Rs.1,500 is fixed **production** cost (so Rs.500 is non-production fixed cost). Normal output 1,500 units per period. No opening stock at the start of Period 1.

Period 1 — Sales 1,200 units, Production 1,500 units. Period 2 — Sales 1,800 units, Production 1,500 units.

Method — the order that keeps you out of trouble:

1. **Step 1: build the stock statement in units first.** Opening + Production – Sales = Closing. P1: $0 + 1,500 - 1,200 = 300$. P2: $300 + 1,500 - 1,800 = 0$.
2. **Step 2: compute the absorption rate** = variable production cost + (fixed production OH ÷ **normal** output) = $4 + (1,500 \div 1,500) = \text{Rs.5 p.u.}$
3. **Step 3: absorption statement** — value both closing and opening stock at Rs.5. Only the Rs.500 non-production fixed cost is deducted below gross profit.
4. **Step 4: marginal statement** — value stock at Rs.4 (variable only) and charge the **whole** Rs.2,000 fixed cost as a period cost.
5. **Step 5: reconcile.** Never leave this out.

W.N.1 — Stock statement (units)

Particulars	Period 1	Period 2
Opening stock	–	300
Add: Production	1,500	1,500
Less: Sales	(1,200)	(1,800)
Closing stock	300	Nil

(i) Statement of Profit under ABSORPTION Costing (Rs.)

Particulars	Period 1	Period 2
Sales (1,200 × 6 / 1,800 × 6)	7,200	10,800

Particulars	Period 1	Period 2
Less: Cost of Goods Sold		
Opening stock (300 × 5)	–	1,500
Add: Production cost (1,500 × 5)	7,500	7,500
Less: Closing stock (300 × 5)	(1,500)	–
<i>Cost of Goods Sold</i>	<i>(6,000)</i>	<i>(9,000)</i>
<i>Gross Profit</i>	<i>1,200</i>	<i>1,800</i>
Less: Other fixed cost (2,000 – 1,500)	(500)	(500)
Net Profit	700	1,300

(ii) Statement of Profit under MARGINAL Costing (Rs.)

Particulars	Period 1	Period 2
Sales	7,200	10,800
Less: Variable cost of sales (units sold × 4)	(4,800)	(7,200)
<i>Contribution</i>	<i>2,400</i>	<i>3,600</i>
Less: Fixed cost (entire 2,000 written off as a period cost)	(2,000)	(2,000)
Net Profit	400	1,600

W.N.2 — Reconciliation of the two profits (Rs.)

Particulars	Period 1	Period 2
Profit as per Marginal Costing	400	1,600
Add: Fixed production OH carried <i>in</i> closing stock (300 × Re.1)	300	–
Less: Fixed production OH released <i>from</i> opening stock (300 × Re.1)	–	(300)
Profit as per Absorption Costing	700	1,300

Answer: Absorption costing — Rs.700 and Rs.1,300. Marginal costing — Rs.400 and Rs.1,600.

Comment to write: In Period 1 stock rose by 300 units, so absorption profit is higher by Rs.300. In Period 2 stock fell by 300 units, so absorption profit is lower by Rs.300. Over the two periods together both methods give the same total profit of Rs.2,000, because opening and closing stock for the two periods taken as a whole is nil.

Q17 Two products — profit under marginal costing at two different sales levels

Given: Mill — SP \$20; DM 8, DL 4, variable production OH 2, variable sales OH 2. Stream — SP \$30; DM 14, DL 2, variable production OH 1, variable sales OH 3. Production 15,000 Mill and 6,000 Stream. Fixed costs for the month: production \$40,000, administration \$15,000, selling & distribution \$25,000.

Method:

1. Add up **all four** variable elements per unit — including **variable sales overhead**, which students routinely omit. Mill = 8+4+2+2 = \$16; Stream = 14+2+1+3 = \$20.
2. Under marginal costing, contribution is based on **units SOLD**, not units produced. Production figures are a distraction here.
3. Total the fixed costs once (40,000 + 15,000 + 25,000 = \$80,000) and deduct from total contribution — never product by product.

(a) Profit for April 20X1 — sales 10,000 Mill and 5,000 Stream (\$)

Particulars	Mill	Stream	Total
Sales (10,000 × 20 / 5,000 × 30)	2,00,000	1,50,000	3,50,000
Less: Variable cost (10,000 × 16 / 5,000 × 20)	(1,60,000)	(1,00,000)	(2,60,000)
<i>Contribution</i>	<i>40,000</i>	<i>50,000</i>	<i>90,000</i>

Particulars	Mill	Stream	Total
Less: Fixed costs (40,000 + 15,000 + 25,000)			(80,000)
Profit			10,000

(b) Profit if sales were 15,000 Mill and 6,000 Stream (\$)

Particulars	Mill	Stream	Total
Sales (15,000 × 20 / 6,000 × 30)	3,00,000	1,80,000	4,80,000
Less: Variable cost (15,000 × 16 / 6,000 × 20)	(2,40,000)	(1,20,000)	(3,60,000)
<i>Contribution</i>	<i>60,000</i>	<i>60,000</i>	<i>1,20,000</i>
Less: Fixed costs			(80,000)
Profit			40,000

Answer: (a) Profit **\$10,000** (b) Profit **\$40,000**.

In (b) the entire production is sold, so there is no closing inventory and marginal and absorption profit would be identical.

Q18 Big Woof Co — quarterly profit statements under both methods

Given: SP \$180; DM 40, DL 16, Variable OH 10 (VC = \$66 p.u.). Annual fixed production OH budgeted \$1,600,000 for a budgeted 1,280,000 units; actual annual fixed OH also \$1,600,000. Budgeted fixed selling cost \$320,000 **per quarter**. Jan-Mar: Sales 240,000 units, Production 280,000 units. No opening inventory.

Method — two conversions must be done before anything else:

- Fixed production OH per unit** = $1,600,000 \div 1,280,000 = \1.25 p.u. (use the **annual** figures for the rate).
- Convert the annual fixed production OH to a quarterly figure** for the profit statement: $1,600,000 \div 4 = \$400,000$. Fixed selling cost is **already** quarterly at \$320,000 — do not divide it again.
- Absorption rate = $66 + 1.25 = \$67.25 \text{ p.u.}$
- Closing stock = $280,000 - 240,000 = 40,000 \text{ units}$.
- In the absorption statement remember the **under/over absorption adjustment**: absorbed = $280,000 \times 1.25 = \$350,000$ against an actual quarterly charge of \$400,000, so \$50,000 is **under-absorbed** and must be deducted.

(a) Statement of Profit or Loss under MARGINAL Costing — quarter ended 31st March 20X8 (\$)

Particulars	\$
Sales (2,40,000 × 180)	4,32,00,000
Less: Variable cost of sales (2,40,000 × 66)	(1,58,40,000)
<i>Contribution</i>	<i>2,73,60,000</i>
Less: Fixed costs	
Fixed production overhead (16,00,000 × 1/4)	(4,00,000)
Fixed selling cost	(3,20,000)
Profit	2,66,40,000

(b) Statement of Profit or Loss under ABSORPTION Costing — quarter ended 31st March 20X8 (\$)

Particulars	\$
Sales (2,40,000 × 180)	4,32,00,000
Less: Cost of Goods Sold	
Opening stock	—
Add: Cost of production (2,80,000 × 67.25)	1,88,30,000

Particulars	\$
Less: Closing stock (40,000 × 67.25)	(26,90,000)
<i>Cost of Goods Sold</i>	<i>(1,61,40,000)</i>
<i>Gross Profit</i>	<i>2,70,60,000</i>
Less: Fixed selling cost	(3,20,000)
<i>Profit before absorption adjustment</i>	<i>2,67,40,000</i>
Less: Under-absorbed fixed production OH (4,00,000 – 3,50,000)	(50,000)
Profit	2,66,90,000

W.N.1 — Reconciliation (\$)

Profit as per Marginal Costing	2,66,40,000
Add: Fixed production OH included in closing stock (40,000 × 1.25)	50,000
Profit as per Absorption Costing	2,66,90,000

Answer: Marginal costing profit **\$2,66,40,000**; Absorption costing profit **\$2,66,90,000**. Absorption profit is higher by \$50,000 because inventory rose by 40,000 units, each carrying \$1.25 of fixed production overhead forward.

Why the answer is still higher despite deducting \$50,000: marginal costing charges the whole \$400,000 quarterly fixed production overhead to this quarter, whereas absorption costing charges only \$350,000 (the amount absorbed) and parks \$50,000 inside closing stock. The two \$50,000 amounts are not the same item — one is the under-absorption, the other is the overhead deferred in stock. Do not "net them off".

Q20 Stoney — three-month comparative statements on both bases

Given: DM \$10, DL \$5, Variable production expenses \$8 (VC = \$23 p.u.). Fixed production cost \$27,000 per month. Normal output 9,000 units per month. SP \$30.
Month 1 — Production 6,500, Sales 5,000. Month 2 — Production 9,000, Sales 8,500. Month 3 — Production 10,000, Sales 9,500.

Method — follow these five steps in order:

1. **Stock statement in units first** (Opening + Production – Sales = Closing), carrying each month's closing forward as the next month's opening.
2. Marginal costing: stock at **\$23**; charge the full \$27,000 fixed cost every month.
3. Absorption rate = $23 + (27,000 \div 9,000) = 23 + 3 = \mathbf{\$26}$. Stock at \$26.
4. Absorption costing: compute **under/over absorption every month** by comparing (production × \$3) with the actual \$27,000.
5. Reconcile month by month.

W.N.1 — Stock statement (units)

Particulars	Month 1	Month 2	Month 3
Opening stock	–	1,500	2,000
Add: Production	6,500	9,000	10,000
Less: Sales	(5,000)	(8,500)	(9,500)
Closing stock	1,500	2,000	2,500

(i) Statement of Profit under MARGINAL Costing (\$)

Particulars	Month 1	Month 2	Month 3
Sales (@ \$30)	1,50,000	2,55,000	2,85,000
Less: Variable cost of sales			

Particulars	Month 1	Month 2	Month 3
Opening stock (@ \$23)	–	34,500	46,000
Add: Cost of production (@ \$23)	1,49,500	2,07,000	2,30,000
Less: Closing stock (@ \$23)	(34,500)	(46,000)	(57,500)
<i>Variable Cost of Sales</i>	<i>(1,15,000)</i>	<i>(1,95,500)</i>	<i>(2,18,500)</i>
<i>Contribution</i>	<i>35,000</i>	<i>59,500</i>	<i>66,500</i>
Less: Fixed production cost	(27,000)	(27,000)	(27,000)
Profit	8,000	32,500	39,500

(ii) Statement of Profit under ABSORPTION Costing (\$) — absorption rate \$26 p.u.

Particulars	Month 1	Month 2	Month 3
Sales (@ \$30)	1,50,000	2,55,000	2,85,000
Less: Cost of Goods Sold			
Opening stock (@ \$26)	–	39,000	52,000
Add: Cost of production (@ \$26)	1,69,000	2,34,000	2,60,000
Less: Closing stock (@ \$26)	(39,000)	(52,000)	(65,000)
<i>Cost of Goods Sold</i>	<i>(1,30,000)</i>	<i>(2,21,000)</i>	<i>(2,47,000)</i>
<i>Gross Profit</i>	<i>20,000</i>	<i>34,000</i>	<i>38,000</i>
Less: Under-absorbed fixed OH	(7,500)	Nil	3,000
Add: Over-absorbed fixed OH (W.N.2)			
Net Profit	12,500	34,000	41,000

W.N.2 — Under / over absorption of fixed production overhead (\$)

Particulars	Month 1	Month 2	Month 3
Overhead absorbed (production × \$3)	19,500	27,000	30,000
Actual fixed overhead incurred	27,000	27,000	27,000
Under (–) / Over (+) absorbed	(7,500)	Nil	+3,000

W.N.3 — Reconciliation Statement (\$)

Particulars	Month 1	Month 2	Month 3
Profit as per Marginal Costing	8,000	32,500	39,500
Add: Fixed OH in closing stock (closing units × \$3)	4,500	6,000	7,500
Less: Fixed OH in opening stock (opening units × \$3)	–	(4,500)	(6,000)
Profit as per Absorption Costing	12,500	34,000	41,000

Answer: Marginal costing — \$8,000, \$32,500, \$39,500. Absorption costing — \$12,500, \$34,000, \$41,000.

Comment: stock rises in every month, so absorption profit exceeds marginal profit throughout. In Month 2 production exactly equals normal output of 9,000 units, so there is neither under- nor over-absorption.

MODULE 3 — Cost-Volume-Profit Analysis (Decision Making)

The four decision rules that cover every problem in this module

- **1. Accept / reject a special or export order.** Fixed cost is already being recovered by existing sales, so judge the order on **contribution alone**. Accept if the order's contribution is positive, i.e. if the offered price exceeds variable cost per unit (including any extra packing, duty or freight for that order). Always add the caution: *accept only if it does not spoil the existing market price and spare capacity exists*.
- **2. Make or buy.** Compare **buying price + unavoidable fixed cost** against the **total cost of making**. Fixed cost that continues either way must be added to the buy option, otherwise the comparison is rigged.
- **3. Key factor / limiting factor (labour hours, machine hours, raw material).** Ranking is by **contribution per unit of the limiting factor**, never by contribution per unit of product. Then allocate the scarce resource to the highest rank first.
- **4. Shut-down decision.** Shut-down point (units) = Avoidable Fixed Cost ÷ Contribution per unit where Avoidable FC = FC if operating – (FC if shut + shut-down costs). If actual output is **below** the shut-down point, close; if **above**, continue; if equal, it is the point of indifference.

Q1 Minimum price for an additional 20% capacity order

Given: Average profit Rs.2.50 on a selling price of Rs.14.50; 60,000 units sold at 60% capacity. Cost per unit — Direct Materials Rs.4, Direct Wages Re.1, Factory OH (variable) Rs.3, Selling OH (variable) Re.0.25. Total fixed cost Rs.2,25,000. Current year: fixed cost +10%, material and labour +5% each. Offer for a further 20% of capacity; required overall profit Rs.1,60,000.

Method — five steps, in this exact order:

1. **Revise the variable cost** for the 5% increases. Only material and labour rise; factory and selling overheads stay put.
2. **Revise the fixed cost:** $2,25,000 \times 1.10$.
3. **Work out the units in the new order.** 60% capacity = 60,000 units, so 1% = 1,000 units and 20% = **20,000 units**. This conversion is worth a mark on its own.
4. **Compute the profit from existing sales** at the revised costs (the old selling price of Rs.14.50 continues for the home market).
5. **Let the new price be x**, form the equation "existing profit + new-order contribution = 1,60,000" and solve.

W.N.1 — Revised variable cost per unit

Element	Original	Revised (Rs.)
Direct Material (4 + 5%)	4.00	4.20
Direct Wages (1 + 5%)	1.00	1.05
Factory overhead (variable) — no change	3.00	3.00
Selling overhead (variable) — no change	0.25	0.25
Total Variable Cost per unit	8.25	8.50

W.N.2 — Revised fixed cost = $2,25,000 + 10\% = \text{Rs.}2,47,500$

W.N.3 — Units in the offer: 60% = 60,000 units $\Rightarrow$ 20% = **20,000 units**

Statement of Profit — evaluation of the new offer

Particulars	Existing market (60,000 units)	New offer (20,000 units)	Total
Selling price per unit	14.50	x	
Less: Variable cost per unit	(8.50)	(8.50)	
<i>Contribution per unit</i>	<i>6.00</i>	<i>x – 8.50</i>	
Total Contribution	3,60,000	$20,000x - 1,70,000$	

Particulars	Existing market (60,000 units)	New offer (20,000 units)	Total
Less: Fixed Cost	(2,47,500)	Nil	
Profit	1,12,500	20,000x – 1,70,000	1,60,000

$$1,12,500 + 20,000x - 1,70,000 = 1,60,000$$

$$20,000x = 1,60,000 + 1,70,000 - 1,12,500 = 2,17,500$$

$$x = 2,17,500 \div 20,000 = \text{Rs.10.875}$$

Answer: The minimum price for the additional 20,000 units is **Rs.10.875 per unit** (say Rs.10.88).

Recommendation to write: Since Rs.10.875 exceeds the variable cost of Rs.8.50, the order contributes Rs. 2.375 per unit and may be accepted at any price at or above Rs.10.875, provided it does not disturb the existing market price of Rs.14.50.

Q2 Three proposals to improve profitability — comparative table

Given: SP Rs.75 p.u.; present sales 40,000 units per month = 50% of capacity; VC Rs.50 p.u.; FC Rs.10 lakhs per month.

I Export 30,000 units at Rs.60 with an extra variable cost of Rs.5 p.u. **II** Domestic chain stores 30,000 units at Rs. 55, existing sales retained. **III** Reduce selling price: by Rs.5 → +10,000 units; by Rs.8 → +30,000 units; by Rs.11 → +35,000 units.

Method:

1. Establish the present position first — it turns out to be exactly break-even (profit nil), which is the benchmark.
2. Proposals I and II are **additional** business, so existing fixed cost is already recovered — their whole contribution is incremental profit.
3. Proposal III replaces the existing price for **all** units, so recompute total sales quantity (40,000 + increase) at the reduced price and deduct the full fixed cost.
4. Present everything in one columnar table and end with a recommendation.

Comparative Statement of Profitability (Rs.)

Particulars	Present	I Export	II Chain stores	III(a) SP –5	III(b) SP –8	III(c) SP –11
Selling price per unit	75	60	55	70	67	64
Less: Variable cost p.u.	(50)	(55)	(50)	(50)	(50)	(50)
<i>Contribution per unit</i>	25	5	5	20	17	14
Number of units	40,000	30,000	30,000	50,000	70,000	75,000
<i>Total Contribution</i>	10,00,000	1,50,000	1,50,000	10,00,000	11,90,000	10,50,000
Less: Fixed Cost	(10,00,000)	Nil	Nil	(10,00,000)	(10,00,000)	(10,00,000)
Profit	Nil	1,50,000	1,50,000	Nil	1,90,000	50,000

Recommendation: Proposal III(b) — reducing the selling price by Rs.8 to Rs.67 and raising volume to 70,000 units — yields the highest monthly profit of **Rs.1,90,000** and is therefore recommended.

Supporting comments: Proposals I and II each add Rs.1,50,000 but the export option also carries the risk of the extra Rs.5 packing/duty cost. Proposal III(a) merely returns the company to break-even, and III(c) sacrifices too much contribution per unit (Rs.14) to be worthwhile. Note that 70,000 units is within the available capacity of 80,000 units.

Q3 Make or buy a component

Given: Cost to make component X-2730 Rs.6.25 (Materials 2.75, Labour 1.75, Other variables 0.50, Depreciation & other fixed 1.25). Market price Rs.4.85 with assured supply.

Method — the whole question turns on one idea:

1. **Depreciation and other fixed costs of Rs.1.25 will be incurred whether or not the component is made.** They are unavoidable. So they must be **added to the buying price** to make a fair comparison.
2. Compare "buy price + unavoidable fixed" against "total cost to make".
3. Choose the lower, then repeat for the revised offer in part (b).

Statement comparing cost of buying with cost of making (per component, Rs.)

Particulars	Buy @ 4.85	Make	Buy @ 5.85
Purchase price	4.85	-	5.85
Materials	-	2.75	-
Labour	-	1.75	-
Other variables	-	0.50	-
Depreciation & other fixed cost (unavoidable — incurred either way)	1.25	1.25	1.25
Total relevant cost	6.10	6.25	7.10

(a) Buying costs Rs.6.10 against Rs.6.25 to make, a saving of Re.0.15 per component. → **The company should BUY.**

(b) At Rs.5.85 the buying cost becomes Rs.7.10 against Rs.6.25 to make, i.e. Rs.0.85 dearer. → **The company should MAKE.**

Non-cost factors worth a line: assurance of supply, quality control, and whether the released capacity can earn a contribution elsewhere.

Q4 Should a foreign order below normal price be accepted?

Given: Capacity 5,000 articles; present output 2,000 for the home market at Rs.80. Costs for 2,000 units — Materials 40,000; Wages 36,000; Factory OH fixed 12,000 and variable 20,000; Administration OH fixed 18,000; Selling & distribution fixed 10,000 and variable 16,000; total Rs.1,52,000. Foreign order for 3,000 articles at Rs.65, with additional packing cost Rs.3,000.

Method:

1. **Separate fixed from variable** in the given cost list. Variable = Materials + Wages + Factory variable + S&D variable. Fixed = 12,000 + 18,000 + 10,000 = Rs.40,000.
2. **Scale the variable costs up by 3,000/2,000 = 1.5x** for the foreign order. Fixed cost stays at Rs.40,000 because total output of 5,000 is still within capacity.
3. Add the special packing cost of Rs.3,000 to the foreign column only.
4. Show both columns and a total, then give the recommendation.

Statement of Profitability (Rs.)

Particulars	Home market 2,000 units	Foreign order 3,000 units	Total 5,000 units
Sales (2,000 × 80 / 3,000 × 65)	1,60,000	1,95,000	3,55,000
Less: Variable costs			
Materials + Wages	(76,000)	(1,14,000)	(1,90,000)
Factory overhead (variable)	(20,000)	(30,000)	(50,000)
Selling & distribution (variable)	(16,000)	(24,000)	(40,000)
Additional packing cost	-	(3,000)	(3,000)
<i>Contribution</i>	<i>48,000</i>	<i>24,000</i>	<i>72,000</i>
Less: Fixed cost (12,000 + 18,000 + 10,000)	(40,000)	Nil	(40,000)
Profit	8,000	24,000	32,000

Recommendation: The foreign order yields a positive contribution of **Rs.24,000** even after the special packing cost, and no additional fixed cost arises because 5,000 units is within the capacity of 5,000. Total profit rises from Rs.8,000 to **Rs.32,000**. → **The order should be ACCEPTED.**

Add the standard caution: acceptable because it is a separate foreign market and will not affect the home selling price of Rs. 80.

The trap in this problem: the variable factory overhead must be scaled to Rs.30,000 ($20,000 \times 1.5$), not left at Rs. 20,000. Leaving it unscaled inflates the contribution to Rs.34,000 and the answer is wrong even though the decision happens to stay the same.

Q5 Export order — three alternatives, choose the best

Given (at 80% capacity): Sales Rs.32,00,000; Direct materials Rs.10,00,000; Direct labour Rs.4,00,000; Variable overheads Rs.2,00,000; Fixed overheads Rs.13,00,000. An export order needing **half the factory capacity** must be taken in full at 10% below normal domestic prices or rejected.

Alt. I Reject and continue at 80%. **Alt. II** Accept, split capacity 50% export / 50% domestic, turning away excess domestic demand. **Alt. III** Increase capacity: (a) buy equipment, raising fixed cost by Rs.1,00,000, and (b) raise the wage rate by $1\frac{1}{2}$ times to get extra output.

Method — get the 100% base first:

- Sales at 100% = $32,00,000 \times 100/80 = \text{Rs.40,00,000}$. So 10% of capacity = Rs.4,00,000 of sales.
- Scale each variable cost the same way, i.e. multiply by (required % ÷ 80).
- Export price is 10% below normal, so export sales = (capacity % of 40,00,000) × 0.90.
- In Alt. III the total requirement is 80% domestic + 50% export = 130%. The extra 30% comes from the new equipment and the last 20% from the higher wage rate, so **labour cost on that 20% slice is $1.5\times$** .

Alternatives I and II (Rs.)

Particulars	Alt. I Domestic 80%	Alt. II Domestic 50%	Alt. II Export 50%	Alt. II Total
Sales	32,00,000	20,00,000	18,00,000	38,00,000
Less: Materials	(10,00,000)	(6,25,000)	(6,25,000)	(12,50,000)
Less: Labour	(4,00,000)	(2,50,000)	(2,50,000)	(5,00,000)
Less: Variable overheads	(2,00,000)	(1,25,000)	(1,25,000)	(2,50,000)
Contribution	16,00,000	10,00,000	8,00,000	18,00,000
Less: Fixed overheads	(13,00,000)			(13,00,000)
Profit	3,00,000			5,00,000

Alternative III — capacity increased to 130% (Rs.)

Particulars	Domestic 80%	Export 30% (new equipment)	Export 20% (wages × $1\frac{1}{2}$)	Total
Sales	32,00,000	10,80,000	7,20,000	50,00,000
Less: Materials	(10,00,000)	(3,75,000)	(2,50,000)	(16,25,000)
Less: Labour	(4,00,000)	(1,50,000)	(1,50,000)	(7,00,000)
Less: Variable overheads	(2,00,000)	(75,000)	(50,000)	(3,25,000)
Contribution	16,00,000	4,80,000	2,70,000	23,50,000
Less: Fixed overheads (13,00,000 + 1,00,000)				(14,00,000)
Profit				9,50,000

Note on the Export 20% slice: normal labour for 20% of capacity = $(20/80) \times 4,00,000 = \text{Rs.1,00,000}$; at $1\frac{1}{2}$ times the wage rate this becomes Rs.1,50,000.

Recommendation: profits are Alt. I Rs.3,00,000, Alt. II Rs.5,00,000 and **Alt. III Rs.9,50,000. Alternative III is the best** — increase capacity, keep the full domestic sales of 80% and execute the entire export order, since the additional fixed cost of Rs.1,00,000 and the higher wage rate are more than covered by the extra contribution of Rs.7,50,000.

Q6 Prestige Company — reduce price and sell more, or raise price and sell less?

Given budget for 20X5-X6: Raw materials 20,00,000; Labour, stores, power and other variable costs 6,00,000; Manufacturing overheads (fixed) 7,00,000; Variable distribution costs 4,00,000; General overheads including selling (fixed) 3,00,000; total Rs.40,00,000. Sales Rs.50,00,000; budgeted profit Rs.10,00,000.

General Manager: cut SP by 5%, volume +50%; extra production-planning cost Rs.50,000 and a new sales office Rs.1,00,000 p.a.

Sales Manager: raise SP by 10%, volume –10%; savings of Rs.50,000 in manufacturing overheads and Rs.1,00,000 in general overheads.

Method — separate variable from fixed before touching anything:

1. Variable costs = Raw materials 20,00,000 + Labour/stores/power 6,00,000 + Variable distribution 4,00,000 = **Rs.30,00,000**. Fixed = 7,00,000 + 3,00,000 = **Rs.10,00,000**.
2. **Sales change twice over** — once for price, once for volume. GM: 50,00,000 × 0.95 × 1.50. SM: 50,00,000 × 1.10 × 0.90.
3. **Variable cost changes only with volume, never with price.** GM: ×1.50. SM: ×0.90. This is the single most common mistake in this problem.
4. Adjust fixed costs for the stated additions or savings, then compare profits.

Comparative Statement of Profitability (Rs.)

Particulars	Present budget	Proposal 1 (GM) SP –5%, vol +50%	Proposal 2 (SM) SP +10%, vol –10%
Sales	50,00,000	71,25,000	49,50,000
Less: Variable costs			
Raw materials	(20,00,000)	(30,00,000)	(18,00,000)
Labour, stores, power & other variable	(6,00,000)	(9,00,000)	(5,40,000)
Variable distribution costs	(4,00,000)	(6,00,000)	(3,60,000)
<i>Total Variable Cost</i>	<i>(30,00,000)</i>	<i>(45,00,000)</i>	<i>(27,00,000)</i>
<i>Contribution</i>	<i>20,00,000</i>	<i>26,25,000</i>	<i>22,50,000</i>
Less: Fixed costs			
Manufacturing overheads	(7,00,000)	(7,00,000)	(6,50,000)
General overheads including selling	(3,00,000)	(3,00,000)	(2,00,000)
Additional costs (planning 50,000 + sales office 1,00,000)	–	(1,50,000)	–
Profit	10,00,000	14,75,000	14,00,000

Recommendation: Proposal 1 (the General Manager's) should be accepted, since it gives a profit of **Rs. 14,75,000** against Rs.14,00,000 under Proposal 2 — higher by Rs.75,000 and Rs.4,75,000 above the present budget.

Balanced comment worth writing: the margin between the two is thin. Proposal 1 needs a 50% jump in volume and pushes P/V ratio down from 40% to 36.8%, so it is the riskier route; Proposal 2 lifts the P/V ratio to 45.5% and needs no extra spending. If the 50% volume increase is not certain, Proposal 2 is the safer choice.

Note on the handwritten notes: the notes conclude "choose Option 2", but their own figures show Proposal 1 at Rs.14,75,000 versus Proposal 2 at Rs.14,00,000. On the profitability criterion asked for in the question, **Proposal 1 is the correct answer.**

Q7 Best sales mixture out of three

Given: Product A — Direct materials Rs.10, Direct wages Rs.3, Sales price Rs.20. Product B — Direct materials Rs.9, Direct wages Rs.2, Sales price Rs.15. Variable expenses are allocated at **100% of direct wages**. Fixed expenses Rs.800.

Mixtures: (i) 1,000 A + 2,000 B (ii) 1,500 A + 1,500 B (iii) 2,000 A + 1,000 B.

Method:

1. Read "variable expenses = 100% of direct wages" carefully — it means **another** Rs.3 for A and Rs.2 for B on top of wages, not that wages *are* the variable expenses.
2. Build the marginal cost card per unit for each product to get contribution per unit.
3. For each mixture, multiply contribution p.u. by units, add the two products, deduct the **same** Rs.800 fixed cost every time.
4. Recommend the mixture with the highest total contribution.

(a) Marginal product cost and contribution per unit (Rs.)

Particulars	Product A	Product B
Selling price	20	15
Less: Marginal (variable) cost		
Direct materials	(10)	(9)
Direct wages	(3)	(2)
Variable expenses (100% of direct wages)	(3)	(2)
<i>Marginal cost per unit</i>	<i>(16)</i>	<i>(13)</i>
Contribution per unit	4	2

(b) Total contribution and profit under each sales mixture (Rs.)

Particulars	Mixture (i)	Mixture (ii)	Mixture (iii)
Units of A × Rs.4	1,000 → 4,000	1,500 → 6,000	2,000 → 8,000
Units of B × Rs.2	2,000 → 4,000	1,500 → 3,000	1,000 → 2,000
<i>Total Contribution</i>	<i>8,000</i>	<i>9,000</i>	<i>10,000</i>
Less: Fixed expenses	(800)	(800)	(800)
Profit	7,200	8,200	9,200

Recommendation: Mixture (iii) — 2,000 units of A and 1,000 units of B — is the best, giving the highest profit of **Rs.9,200**.

Reason to write: Product A earns twice the contribution per unit of B (Rs.4 vs Rs.2), so any mix weighted towards A raises total contribution while fixed cost stays unchanged at Rs.800.

Q8 Budgeted profit, then the optimal mix when raw material is restricted

Given: Raw materials at Rs.20 per kg. Per unit — A: RM 80, wages 5, variable OH 10, fixed OH 9, SP 140. B: RM 40, wages 15, variable OH 30, fixed OH 22, SP 120. C: RM 20, wages 10, variable OH 20, fixed OH 18, SP 90. Budgeted production A 6,400, B 3,200, C 2,400 units. Part (ii): raw material restricted to 18,400 kg.

Method for part (ii) — the four-step limiting factor drill:

1. **Convert the material cost into kilograms:** kg per unit = RM cost per unit ÷ Rs.20. A = 4 kg, B = 2 kg, C = 1 kg.
2. **Check that a shortage actually exists:** total requirement = 6,400×4 + 3,200×2 + 2,400×1 = 34,400 kg against 18,400 kg available. Shortage confirmed — state this.
3. **Rank by contribution per kg** (not contribution per unit).
4. **Allocate the 18,400 kg** to rank 1 first, up to its maximum demand, then rank 2, and give the balance to rank 3.

(i) Statement of Budgeted Profit (Rs.)

Particulars	A	B	C	Total
Selling price per unit	140	120	90	
Less: Variable cost p.u. (RM + wages + variable OH)	(95)	(85)	(50)	
<i>Contribution per unit</i>	45	35	40	
Budgeted production (units)	6,400	3,200	2,400	
<i>Total Contribution</i>	2,88,000	1,12,000	96,000	4,96,000
Less: Fixed overheads (9 / 22 / 18 × units)	(57,600)	(70,400)	(43,200)	(1,71,200)
Profit	2,30,400	41,600	52,800	3,24,800

W.N.1 — Ranking by contribution per kg of raw material

Particulars	A	B	C
Contribution per unit (Rs.)	45	35	40
Raw material per unit (kg) (RM cost ÷ 20)	4	2	1
<i>Contribution per kg (Rs.)</i>	11.25	17.50	40.00
Ranking	III	II	I

W.N.2 — Allocation of the 18,400 kg available

Allocation in order of ranking	Units	kg per unit	kg used	Balance kg
Material available				18,400
To C (rank I) — full demand	2,400	1	2,400	16,000
To B (rank II) — full demand	3,200	2	6,400	9,600
To A (rank III) — balance 9,600 ÷ 4	2,400	4	9,600	Nil

(ii) Statement of Profit under the Optimal Product Mix (Rs.)

Particulars	A	B	C	Total
Contribution per unit	45	35	40	
Optimal units	2,400	3,200	2,400	
<i>Total Contribution</i>	1,08,000	1,12,000	96,000	3,16,000
Less: Fixed overheads (unchanged)	(57,600)	(70,400)	(43,200)	(1,71,200)
Profit	50,400	41,600	52,800	1,44,800

Answers: (i) Budgeted profit **Rs.3,24,800**. (ii) Optimal mix **A 2,400 units, B 3,200 units, C 2,400 units**, giving a profit of **Rs.1,44,800**.

Comment: production of A has to be cut from 6,400 to 2,400 units because A earns the lowest contribution per kg (Rs.11.25). Fixed overheads are **not** reduced — they are a period cost and continue in full.

Q9 Profitability of two products during a labour shortage

Given: Product A — SP Rs.75, Direct materials Rs.30, Direct labour 15 hours at Re.0.50 per hour. Product B — SP Rs.48, Direct materials Rs.30, Direct labour 2 hours at Re.0.50. Variable overhead = 100% of direct wages. Fixed overheads Rs.3,000.

Method:

1. Convert labour hours into wages: A = $15 \times 0.50 = \text{Rs.}7.50$; B = $2 \times 0.50 = \text{Re.}1$.
2. Variable overhead equals the same amount again (100% of wages).
3. Get contribution per unit, then divide by **labour hours** — labour is the scarce resource named in the question.
4. Rank on contribution per labour hour and state the rule.

Statement of Profitability per Labour Hour (Rs.)

Particulars	Product A	Product B
Selling price per unit	75	48
Less: Variable cost		
Direct materials	(30)	(30)
Direct wages (15 hrs $\times$ 0.50 / 2 hrs $\times$ 0.50)	(7.50)	(1)
Variable overhead (100% of wages)	(7.50)	(1)
<i>Total variable cost</i>	<i>(45)</i>	<i>(32)</i>
<i>Contribution per unit</i>	<i>30</i>	<i>16</i>
Direct labour hours per unit	15	2
<i>Contribution per labour hour</i>	<i>2</i>	<i>8</i>
Ranking	II	I

Conclusion: Product B is more profitable during a labour shortage, earning Rs.8 of contribution per labour hour against only Rs.2 for A.

Rule to state: when labour is the key factor, the firm should give priority to the product yielding the **maximum contribution per labour hour**, not the maximum contribution per unit. Judged on contribution per unit alone A (Rs.30) looks better than B (Rs.16), and that is precisely the trap.

Q10 Key factor — which product is more profitable, plus the optimal mix in 4,200 hours

Given: Product A — Materials Rs.5, Labour 6 hrs at Re.0.50 = Rs.3, Fixed OH 50% of labour = Rs.1.50, Variable OH Rs.1.50, total cost Rs.11, SP Rs.14, production 500 units. Product B — Materials Rs.5, Labour 3 hrs at Re.0.50 = Rs.1.50, Fixed OH Rs.0.75, Variable OH Rs.1.50, total cost Rs.8.75, SP Rs.11, production 600 units. Maximum labour hours per month 4,200.

Method:

1. Strip out the **fixed** overhead from the total cost — variable cost is Materials + Labour + Variable OH only. A = $5+3+1.50 = \text{Rs.}9.50$; B = $5+1.50+1.50 = \text{Rs.}8$.
2. Prepare the statement of profit at the present output.
3. Rank on contribution per labour hour.
4. Allocate 4,200 hours to the higher rank first, then the balance to the other product.

Statement of Profit at present output (Rs.)

Particulars	Product A	Product B	Total
Selling price per unit	14	11	
Less: Variable cost per unit			
Materials	(5.00)	(5.00)	

Particulars	Product A	Product B	Total
Labour	(3.00)	(1.50)	
Variable overhead	(1.50)	(1.50)	
<i>Total variable cost per unit</i>	<i>(9.50)</i>	<i>(8.00)</i>	
<i>Contribution per unit</i>	<i>4.50</i>	<i>3.00</i>	
Number of units	500	600	
<i>Total Contribution</i>	<i>2,250</i>	<i>1,800</i>	<i>4,050</i>
Less: Fixed cost (1.50 × 500 / 0.75 × 600)	(750)	(450)	(1,200)
Profit	1,500	1,350	2,850

W.N.1 — Ranking of products (labour is the key factor)

Particulars	Product A	Product B
Contribution per unit (Rs.)	4.50	3.00
Labour hours per unit	6	3
<i>Contribution per labour hour (Rs.)</i>	<i>0.75</i>	<i>1.00</i>
Ranking	II	I

W.N.2 — Optimal product mix within 4,200 labour hours

Allocation	Units	Hours per unit	Hours used	Balance hours
Hours available				4,200
Product B (rank I) — full demand	600	3	1,800	2,400
Product A (rank II) — balance 2,400 ÷ 6	400	6	2,400	Nil

Profitability under the optimal mix (Rs.)

Particulars	Product A	Product B	Total
Contribution per unit	4.50	3.00	
Optimal units	400	600	
<i>Total Contribution</i>	<i>1,800</i>	<i>1,800</i>	<i>3,600</i>
Less: Fixed cost			(1,200)
Profit			2,400

Answers: Product B is more profitable (Re.1.00 contribution per labour hour against Re.0.75 for A). Within 4,200 hours the optimal mix is **600 units of B and 400 units of A**, giving a profit of **Rs.2,400**.

Q11 Foreign order at Rs.36 — accept or not? And if it were from a local merchant?

Given per unit: Materials Rs.12, Labour Rs.9, Variable expenses Rs.6, Fixed expenses Rs.18, total Rs.45; sold at Rs.51. Normal capacity 1,00,000 units; the above figures relate to 80,000 units. Offer for 20,000 units at Rs.36 from a foreign customer.

Method:

- Variable cost per unit = 12 + 9 + 6 = **Rs.27**. Fixed cost in total = 18 × 80,000 = **Rs.14,40,000** (a total, not a per-unit cost to be charged to the new order).
- The 20,000 units bring output to 1,00,000, exactly normal capacity — so **no additional fixed cost** arises. Say this explicitly.
- The offered price of Rs.36 is below the total cost of Rs.45 but well above the variable cost of Rs.27, so contribution is Rs.9 per unit.
- Answer the second part on **market** grounds, not arithmetic.

Statement of Profitability (Rs.)

Particulars	Local market 80,000 units	Foreign order 20,000 units	Total 1,00,000 units
Sales (80,000 × 51 / 20,000 × 36)	40,80,000	7,20,000	48,00,000
Less: Variable cost (@ Rs.27 = 12+9+6)	(21,60,000)	(5,40,000)	(27,00,000)
Contribution	19,20,000	1,80,000	21,00,000
Less: Fixed cost (18 × 80,000 — no increase)	(14,40,000)	Nil	(14,40,000)
Profit	4,80,000	1,80,000	6,60,000

Advice on the foreign order: ACCEPT. The order earns a contribution of Rs.9 per unit (Rs.1,80,000 in total) and no extra fixed cost is incurred because output stays within the normal capacity of 1,00,000 units. Total profit rises from Rs.4,80,000 to **Rs.6,60,000**.

Advice if the order came from a local merchant: REJECT. Although the arithmetic is identical, selling at Rs.36 in the **same** market where the product fetches Rs.51 would force the company to cut the price to its existing customers, cause resentment among regular dealers, and destroy far more contribution than the Rs. 1,80,000 gained. A foreign market is geographically separate, so the two prices can coexist.

Q12 What must variable cost per unit be to justify a shut-down?

Given: SP \$14 p.u.; demand for the period only 5,000 units; fixed expenses at 50% activity (5,000 units) \$30,000. On shut-down an additional \$2,000 must be spent and only \$20,000 of the fixed costs can be avoided.

Method — set up the two alternatives and equate them:

1. **Continue:** Profit = Sales – (5,000 × VC p.u.) – Fixed cost. Let VC per unit = x.
2. **Shut down:** there is no sales revenue, but unavoidable fixed cost of (30,000 – 20,000) = \$10,000 plus shut-down cost \$2,000 = a loss of **\$12,000**.
3. **Equate the two** to find the point of indifference. Above that variable cost, shutting down is better.

Comparison of the two alternatives (\$)

Particulars	Continue operations	Shut down
Sales (5,000 × 14)	70,000	Nil
Less: Variable cost (5,000x)	(5,000x)	Nil
Contribution	70,000 – 5,000x	Nil
Less: Fixed cost (unavoidable 10,000 + shut-down 2,000)	(30,000)	(12,000)
Profit / (Loss)	40,000 – 5,000x	(12,000)

At the point of indifference the two outcomes are equal:

$$40,000 - 5,000x = -12,000$$

$$5,000x = 40,000 + 12,000 = 52,000$$

$$x = 52,000 \div 5,000 = \mathbf{\$10.40}$$

Answer: The variable cost per unit would have to be **\$10.40** for the company to be indifferent. **A shut-down should be recommended only if the variable cost per unit exceeds \$10.40**, because beyond that the loss from operating is greater than the \$12,000 loss from closing.

Rule to write: Output > shut-down point → continue; Output = shut-down point → point of indifference; Output < shut-down point → close operations.

Q13 G Ltd — should the plant be shut down? What is the shut-down capacity level?

Given: 95,000 units of X sold in a year at 80% capacity; SP \$8 p.u.; variable cost 75% of selling price; fixed cost \$350,000. On shut-down, fixed cost falls to \$130,000 and additional shut-down costs of \$15,000 arise.

Method — four lines and you are done:

1. Contribution per unit = $8 - (75\% \text{ of } 8) = 8 - 6 = \2 .
2. **Avoidable fixed cost** = FC if operating – (FC if shut + shut-down costs) = $350,000 - (130,000 + 15,000) = \$205,000$.
3. Shut-down point = Avoidable FC ÷ Contribution p.u.
4. Compare with actual output and convert to a capacity percentage.

Step	Working	Result
Contribution per unit	$8 - 6$	\$2
Fixed cost if operations continue	given	\$3,50,000
Fixed cost if shut down	$1,30,000 + 15,000$	\$1,45,000
Avoidable fixed cost	$3,50,000 - 1,45,000$	\$2,05,000
Shut-down point (units)	$2,05,000 \div 2$	1,02,500 units

Capacity level at the shut-down point

$95,000 \text{ units} = 80\% \text{ capacity} \Rightarrow 100\% \text{ capacity} = 95,000 \times 100/80 = 1,18,750 \text{ units}$

Shut-down capacity level = $(1,02,500 \div 1,18,750) \times 100 = 86.32\%$

Answer: The shut-down point is **1,02,500 units (86.32% of capacity)**. Present output is only 95,000 units, which is **below** the shut-down point, so **the plant should be shut down**.

Proof to include: if operations continue, loss = $(95,000 \times 2) - 3,50,000 = 1,90,000 - 3,50,000 = \text{loss of } \$1,60,000$. If shut down, loss = **\$1,45,000**. Closing therefore saves \$15,000.

Q14 Paint company — shut down for the quarter? And the shut-down point in tins

Given: Normal output 200,000 tins a year; cost per tin \$16.40 — Direct materials 7.80, Direct labour 2.10, Variable OH 2.50, Fixed OH 4.00. SP \$21 per tin with variable selling expenses of \$0.60 per tin. Next quarter only 10,000 tins can be produced and sold. Fixed manufacturing cost can be cut to \$74,000 for the quarter on shut-down; additional shut-down costs \$14,000.

Method — mind the annual-to-quarterly conversion:

1. Contribution per tin = $21 - (7.80 + 2.10 + 2.50 + 0.60) = 21 - 13 = \8 . Remember to include the variable **selling** expense of \$0.60.
2. Fixed OH for the **year** = $4.00 \times 200,000 = \$800,000$, so for the **quarter** = $\$800,000 \div 4 = \$200,000$. (The question says fixed overheads accrue at a uniform rate through the year.)
3. Fixed cost if shut = $74,000 + 14,000 = \$88,000$.
4. Avoidable FC = $200,000 - 88,000 = \$112,000$; shut-down point = $112,000 \div 8$.

Comparison for the quarter (\$)

Particulars	Continue	Shut down
Contribution (10,000 tins × \$8)	80,000	Nil
Less: Fixed manufacturing cost for the quarter	(2,00,000)	(74,000)
Less: Additional shut-down costs	-	(14,000)
Loss	(1,20,000)	(88,000)

Avoidable fixed cost = $2,00,000 - 88,000 = \$1,12,000$

Shut-down point = $1,12,000 \div 8 = 14,000 \text{ tins}$

(a) Opinion: the activity level of 10,000 tins is **below** the shut-down point of 14,000 tins, and closing reduces the loss from \$1,20,000 to \$88,000. → **The plant should be shut down for the quarter**, saving \$32,000.

(b) Shut-down point = 14,000 tins. Above 14,000 tins operations should continue, because the contribution then exceeds the avoidable fixed cost.

Worth adding: non-cost factors such as loss of trained labour, customer goodwill and re-start costs should also be weighed before actually closing.

Q15 Prepare a break-even chart

Given: Fixed Cost Rs.2,000; Variable Cost Re.0.50 per unit; Sales Re.1 per unit; units 2,000 / 4,000 / 6,000 / 8,000 / 10,000.

Method for any break-even chart question:

1. **Always prepare the computation table first**, then draw. The table itself carries marks.
2. Verify the BEP by formula so the graph can be checked: $BEP = FC \div \text{Contribution p.u.} = 2,000 \div 0.50 = \mathbf{4,000 \text{ units}}$.
3. On the graph: **X-axis = output in units, Y-axis = cost and revenue in Rs.** Draw (a) fixed cost line horizontal at Rs.2,000, (b) total cost line starting at Rs.2,000 on the Y-axis, (c) sales/revenue line from the origin.
4. Mark the intersection of the sales line and total cost line as the **Break-Even Point**; label the loss area to the left and the profit area to the right, and show the margin of safety.

Computation of profit at various levels of output (Rs.)

Particulars	2,000 u	4,000 u	6,000 u	8,000 u	10,000 u
Sales (@ Re.1)	2,000	4,000	6,000	8,000	10,000
Less: Variable cost (@ Re.0.50)	(1,000)	(2,000)	(3,000)	(4,000)	(5,000)
Contribution	1,000	2,000	3,000	4,000	5,000
Less: Fixed cost	(2,000)	(2,000)	(2,000)	(2,000)	(2,000)
Profit / (Loss)	(1,000)	Nil	1,000	2,000	3,000

Break-Even Point = 4,000 units = Rs.4,000 of sales, where the profit column is exactly nil — the chart must show the sales line cutting the total cost line at this point.

Total cost line values to plot: 3,000 / 4,000 / 5,000 / 6,000 / 7,000 at the five output levels (fixed 2,000 + variable).

Q16 Break-even chart, and the new BEP if the selling price falls to Rs.18

Given: Budgeted output 1,00,000 units; Fixed expenses Rs.5,00,000; Variable expenses Rs.10 per unit; Selling price Rs.20 per unit.

Method:

1. Contribution p.u. = $20 - 10 = \text{Rs.}10$; $BEP = 5,00,000 \div 10 = \mathbf{50,000 \text{ units}}$.
2. Tabulate several output levels around the BEP so the chart can be plotted.
3. For the revised price, **only contribution changes**: $18 - 10 = \text{Rs.}8$. Fixed cost and variable cost are untouched.

Computation at various levels of output (Rs. in lakhs unless stated)

Particulars	50,000 u	1,00,000 u	1,50,000 u	2,00,000 u	2,50,000 u
Sales (@ Rs.20)	10.00	20.00	30.00	40.00	50.00
Less: Variable cost (@ Rs.10)	(5.00)	(10.00)	(15.00)	(20.00)	(25.00)
Contribution	5.00	10.00	15.00	20.00	25.00
Less: Fixed cost	(5.00)	(5.00)	(5.00)	(5.00)	(5.00)
Profit / (Loss)	Nil	5.00	10.00	15.00	20.00

Existing BEP = $5,00,000 \div 10 = \mathbf{50,000 \text{ units}}$ (= Rs.10,00,000 of sales), confirmed by the nil profit column.

New BEP if SP falls to Rs.18: revised contribution = $18 - 10 = \text{Rs.}8 \text{ p.u.}$, so $BEP = 5,00,000 \div 8 = \mathbf{62,500 \text{ units}}$ (= Rs.11,25,000 of sales).

Comment: a 10% cut in selling price raises the break-even volume by 25%, because contribution per unit falls by 20% while fixed cost stays the same. This shows how sensitive the BEP is to price.

MODULE 4 — Relevant Costing

The decision tree for materials — draw this in the margin before you start any relevant cost problem

- **Is the material in stock?**
 - **NO** → it has to be bought → relevant cost = **out-of-pocket cost** = **current purchase / replacement price**.
 - **YES** → ask the next question:
 - **Is it used regularly? YES** → using it now means replacing it later → relevant cost = **replacement cost**.
 - **NO** (non-moving / surplus stock) → relevant cost = **opportunity cost**, being the higher of its **scrap value / net realisable value** or the **saving from using it as a substitute** for some other material.

Relevant vs Irrelevant — the list you must be able to recite

- **RELEVANT:** future cash flows, marginal / variable cost, differential (incremental) cost, opportunity cost, replacement cost, out-of-pocket cost, avoidable cost, imputed cost, **specific** fixed cost incurred only for this decision.
- **IRRELEVANT:** sunk cost (already incurred — e.g. book value, original purchase price), committed cost, **apportioned / absorbed** fixed overhead, non-cash items such as depreciation, and the cost of permanent labour that will be paid anyway.
- **The two golden tests:** (1) does it change **in the future** as a result of this decision? (2) does it involve a **cash flow**? If the answer to either is no, ignore the item — but **write one line saying why you ignored it**, because that sentence carries marks.

Q2 Relevant cost of materials for a special job offered at Rs.22,000

Given: Material A — need 1,000, in stock 0, replacement Rs.6 p.u. Material B — need 1,000, in stock 600 (book Rs. 2, realisable Rs.2.50), replacement Rs.5 p.u.; **used regularly**. Material C — need 1,000, in stock 700 (book Rs.3, realisable Rs.2.50), replacement Rs.4 p.u.; no other use. Material D — need 200, in stock 200 (book Rs.4, realisable Rs.6), replacement Rs.9 p.u.; could instead substitute for 300 units of Material E which costs Rs.5 p.u. and is not in stock.

Method — take each material through the decision tree separately:

1. **A** — nothing in stock, so all 1,000 units must be bought at Rs.6. Straight out-of-pocket cost.
2. **B** — used regularly, so whether we take it from stock or buy it, the company ends up buying 1,000 units. Value the **whole 1,000** at replacement cost Rs.5. Book value Rs.2 and realisable Rs.2.50 are both irrelevant.
3. **C** — the 700 in stock have **no other use**, so their opportunity cost is only their realisable value of Rs.2.50. The remaining 300 must be bought at Rs.4.
4. **D** — all 200 are in stock and there are two possible sacrifices: sell them for Rs.6 each (= Rs.1,200) **or** use them to replace 300 units of E costing Rs.5 each (= Rs.1,500). The opportunity cost is the **higher** benefit forgone, i.e. **Rs.1,500**.

Statement of Relevant Cost of Materials

Material	Nature & computation	Amount (Rs.)
A	1,000 units to be purchased. Out-of-pocket cost is the relevant cost — $1,000 \times \text{Rs.6}$	6,000
B	Used regularly, so stock consumed must be replaced. Entire 1,000 units at replacement cost — $1,000 \times \text{Rs.5}$	5,000
C	700 units in stock with no alternative use → opportunity cost = NRV — $700 \times \text{Rs.2.50}$	1,750
	Balance 300 units to be purchased at replacement cost — $300 \times \text{Rs.4}$	1,200
D	200 units in stock. Best alternative use is as a substitute for 300 units of E — $300 \times \text{Rs.5} = \text{Rs.1,500}$, which exceeds the NRV of $200 \times \text{Rs.6} = \text{Rs.1,200}$. Take the higher.	1,500
TOTAL RELEVANT COST OF MATERIALS		15,450

Decision: the customer will pay Rs.22,000 while the relevant material cost is Rs.15,450, leaving **Rs.6,550** to cover labour, overheads and profit. **The order can be accepted provided all other relevant costs together do not exceed Rs.6,550.**

Note to add: all book values (Rs.2, Rs.3, Rs.4) are **sunk costs** and have been ignored.

Q3 Relevant cost of using a specialised machine on a six-month contract

Given: Machine bought 4 years ago for Rs.90,000; present net book value Rs.35,000. The company was about to sell it for Rs.40,000, but if used on this contract it can be sold after 6 months for Rs.25,000. Variable cost of operating it for 6 months is Rs.60,000. Interest cost is ignored.

Method:

1. **Ignore** the original cost of Rs.90,000 and the net book value of Rs.35,000 — both are **sunk costs**. Say so in one line.
2. The real sacrifice from using the machine is the **fall in its resale value**: it could fetch Rs.40,000 today but only Rs.25,000 after the contract. That Rs.15,000 drop is an opportunity cost.
3. Add the out-of-pocket operating cost of Rs.60,000.

Computation of Relevant Cost of the Machine

Nature & computation	Amount (Rs.)
Variable cost of operating the machine for 6 months — out-of-pocket, hence relevant	60,000
Reduction in net realisable value as opportunity cost — (40,000 – 25,000)	15,000
TOTAL RELEVANT COST	75,000

Answer: Relevant cost of using the machine on the contract = **Rs.75,000**. This is the minimum the contract must recover in respect of the machine.

Items ignored, with reasons: original cost Rs.90,000 — sunk; net book value Rs.35,000 — sunk and merely an accounting figure; interest — excluded by the question.

Q4 Noida Camera Company — discuss the relevance of three items in a special order

Given: The company has excess capacity and the special order will not displace regular production. (a) Equipment has a book value of Rs.2,000 and no other use; it would fetch Rs.1,500 if the order is refused, or Rs.800 if used on the order and sold after three months. (b) The order needs storage space; renting space nearby costs Rs.18,000, while building depreciation allocated to that space is Rs.12,000. (c) A sub-assembly can be bought for Rs.24 per unit or made for Rs.30 (Direct material 10, Direct labour 6, Variable overheads 6, Apportioned fixed overhead 8).

Computation of Relevant Cost of the Special Order

Item	Nature & treatment	Relevant amount
(a) Equipment	Book value Rs.2,000 is a sunk cost — irrelevant . The sacrifice is the drop in resale value caused by using it on the order: Rs.1,500 – Rs.800. This reduction in NRV is an opportunity cost and is relevant .	Rs.700
(b) Storage space	Rent of Rs.18,000 for the nearby warehouse is a genuine future cash outflow caused by the order — out-of-pocket, relevant . Building depreciation of Rs.12,000 is a non-cash, merely apportioned charge that continues whether or not the order is taken — irrelevant .	Rs.18,000
(c) Sub-assembly	Only the variable elements are relevant to making: 10 + 6 + 6 = Rs.22 per unit . The apportioned fixed overhead of Rs.8 is irrelevant because it is absorbed at a normal rate and does not change with this decision. Making at Rs.22 is cheaper than buying at Rs.24, so make in-house and save Rs.2 per unit.	Rs.22 p.u.

Conclusion: the relevant costs are the Rs.700 fall in equipment value, the Rs.18,000 warehouse rent, and Rs. 22 per unit for the sub-assembly which should be **made, not bought**. Book value, allocated building depreciation and apportioned fixed overhead are all excluded as irrelevant.

Q5 Compute the opportunity cost of each of three mutually exclusive products

Given: Only one of X, Y or Z can be made in a year. X — SP Rs.10, VC Rs.6, demand 3,000, capacity 2,000. Y — SP Rs.12, VC Rs.9, demand 2,000, capacity 3,000. Z — SP Rs.12, VC Rs.7, demand 1,000, capacity 900. Fixed cost Rs. 30,000.

Method — two ideas do all the work:

- Effective production = the LOWER of demand and capacity.** You cannot make more than capacity, and you cannot sell more than demand. X → 2,000; Y → 2,000; Z → 900.
- Opportunity cost of choosing a product = the contribution of the BEST alternative forgone** (not the total of all others). Fixed cost of Rs.30,000 is common to all three and therefore irrelevant to the comparison.

Computation of Opportunity Cost

Particulars	X	Y	Z
Selling price per unit (Rs.)	10	12	12
Less: Variable cost per unit	(6)	(9)	(7)
<i>Contribution per unit</i>	4	3	5
Market demand (units)	3,000	2,000	1,000
Production capacity (units)	2,000	3,000	900
Effective production (lower of the two)	2,000	2,000	900
<i>Total Contribution (Rs.)</i>	8,000	6,000	4,500
Opportunity Cost (best contribution forgone)	6,000	8,000	8,000

Answer: Opportunity cost — X **Rs.6,000** (best forgone is Y at Rs.6,000), Y **Rs.8,000** (best forgone is X), Z **Rs. 8,000** (best forgone is X).

Decision: Product X should be selected, since it yields the highest contribution of Rs.8,000 and carries the lowest opportunity cost of Rs.6,000. Profit = 8,000 – 30,000 fixed = a loss of Rs.22,000, but X still minimises the loss.

Q6 Make the part, buy and use the machine elsewhere, or buy and hire the machine out?

Given: A machine makes 10,000 units of a part at a total cost of Rs.21 each, of which Rs.18 is variable. The part is available in the market at Rs.19. If bought, the machine can either make a component of the same quantity contributing Rs.2 each, or be hired out for Rs.21,000.

Method:

- Use only the **variable** cost of Rs.18 for the make option — the Rs.3 fixed element (21 – 18) continues whichever course is taken, so it can be left out of all three columns.
- Buying costs 10,000 × Rs.19 = Rs.1,90,000 under both "buy" options.
- Deduct the **benefit** the released machine earns, then compare **net relevant cost** — the lowest wins.

Statement of Net Relevant Cost (10,000 units)

Particulars	Make	Buy & make the component	Buy & hire the machine out
Variable cost of making (10,000 × 18)	1,80,000	–	–
Purchase price (10,000 × 19)	–	1,90,000	1,90,000
<i>Gross relevant cost</i>	<i>1,80,000</i>	<i>1,90,000</i>	<i>1,90,000</i>

Particulars	Make	Buy & make the component	Buy & hire the machine out
Less: Relevant benefit from the released machine component $10,000 \times \text{Rs.}2$ / hire charges	Nil	(20,000)	(21,000)
NET RELEVANT COST	1,80,000	1,70,000	1,69,000

Recommendation: Buy the part from the market and hire the machine out — this gives the lowest net relevant cost of **Rs.1,69,000**, saving Rs.11,000 against making the part and Rs.1,000 against using the machine for the other component.

Note: the fixed cost of Rs.3 per unit (Rs.30,000 in total) is unavoidable and has been excluded from all three alternatives, as it does not affect the comparison.

Q7 Minimum price for converting a non-moving material

Given: Material M — book value Rs.60, realisable value Rs.80, replacement cost Rs.100. Converting one unit of M into one unit of finished product needs 1 labour hour; labour is paid Rs.20 per hour. Out-of-pocket expenses Rs.30 per unit; allocated overheads Rs.10 per unit. The labour will be redeployed from other activities with **no loss of contribution**, and the employees are **permanent**.

Method — test each of the five items:

- Material M** is a **non-moving** item, so it is not replaced. Relevant cost = opportunity cost = its **realisable value of Rs.80**. Book value Rs.60 is sunk; replacement cost Rs.100 does not apply because the material will not be replaced.
- Labour** — permanent employees, redeployed, and **no contribution is lost**. The wages are a **committed cost** payable anyway → **irrelevant, Nil**.
- Out-of-pocket expenses Rs.30** — a future cash outflow caused by the order → **relevant**.
- Allocated overheads Rs.10** — merely apportioned, no extra cash → **irrelevant**.

Computation of Minimum Price per Unit

Particulars	Nature & computation	Amount (Rs.)
Material M	Non-moving item, hence opportunity cost = net realisable value	80
Labour	Permanent employment is a committed cost and redeployment causes no loss of contribution — hence irrelevant	Nil
Out-of-pocket expenses	Future cash outflow — relevant	30
Allocated overheads	Apportioned only, no additional cash outflow — irrelevant	Nil
MINIMUM PRICE TO BE CHARGED		110

Answer: The minimum price is **Rs.110 per unit**. At this price the company is no worse off than if it had simply scrapped Material M for Rs.80 and not taken the order.

Add: any price above Rs.110 leaves the company better off; below Rs.110 the order should be refused.

Q8 Replace the machine? Hours required and break-even hours of each machine

Given: A machine was purchased for Rs.20,00,000 with an operating cost of Rs.100 per hour. Before installation, a more efficient machine became available at Rs.35,00,000 with an operating cost of Rs.70 per hour. If the new machine is bought, the old one can be disposed of for Rs.12,00,000. Revenue from either machine is Rs.120 per hour.

Method:

1. **(i) Identify what is relevant.** The Rs.20,00,000 already spent on the old machine is a **sunk cost**. What is relevant: the Rs.35,00,000 outlay, the Rs.12,00,000 disposal proceeds, the two operating costs, and the revenue per hour.
2. **(ii) Net incremental investment** = 35,00,000 – 12,00,000 = **Rs.23,00,000**. The new machine earns a contribution of (120 – 70) = **Rs.50 per hour**. Hours needed = net investment ÷ contribution per hour.
3. **(iii) Break-even hours** for each machine = its own cost ÷ its own contribution per hour.

(i) Statement of Relevant Costs

Item	Treatment
Purchase price of old machine Rs. 20,00,000	Sunk cost — IRRELEVANT for the replacement decision (relevant only for computing that machine's own break-even hours)
Cost of new machine Rs.35,00,000	Future cash outflow — relevant
Disposal value of old machine Rs. 12,00,000	Future cash inflow that arises only if replacement happens — relevant
Operating cost Rs.100 / hr (old) vs Rs.70 / hr (new)	Differential cost of Rs.30 per hour — relevant
Revenue Rs.120 per hour	Relevant , though common to both machines

(ii) Hours the new machine must run to justify the change

Step	Working	Result
Cost of new machine	given	35,00,000
Less: Disposal value of the old machine	given	(12,00,000)
<i>Net relevant investment</i>		<i>23,00,000</i>
Contribution per hour of the new machine	120 – 70	50
Hours required	23,00,000 ÷ 50	46,000 hours

(iii) Break-even hours of each machine

Machine	Cost (Rs.)	Contribution per hour	Break-even hours
Machine 1 (old)	20,00,000	120 – 100 = 20	1,00,000 hours
Machine 2 (new)	35,00,000	120 – 70 = 50	70,000 hours

Answers: (i) The Rs.20,00,000 spent on the old machine is sunk and irrelevant; relevant items are the Rs. 35,00,000 outlay, the Rs.12,00,000 disposal value and the Rs.30 per hour operating saving. (ii) The new machine must run **46,000 hours** to recover the net investment of Rs.23,00,000. (iii) Break-even hours — old machine **1,00,000 hours**, new machine **70,000 hours**.

Recommendation: the new machine breaks even in 30,000 fewer hours and earns Rs.30 per hour more contribution, so **the replacement is worthwhile** provided the expected usage exceeds 46,000 hours.

MODULE 5 — Budget and Budgetary Control

The four rules that carry you through every budget problem

- **1. Cost behaviour (write this at the top of your answer):**
Fixed cost — the **total stays the same**, the **per-unit figure changes**.
Variable cost — the **total changes**, the **per-unit figure stays the same**.
Semi-variable cost — must be **split** into its fixed and variable parts before use.
- **2. Segregating a semi-variable cost** (high-low method):

$$\text{Variable rate per unit} = (\text{Change in total cost}) \div (\text{Change in volume})$$
then
$$\text{Fixed portion} = \text{Total cost at any level} - (\text{Variable rate} \times \text{volume at that level})$$
- **3. Production Budget:**
$$\text{Budgeted Production} = \text{Budgeted Sales} + \text{Desired Closing Stock} - \text{Opening Stock}$$
- **4. Purchase / Procurement Budget:**

$$\text{Materials to be Purchased} = \text{Materials Consumed} + \text{Desired Closing Stock} - \text{Opening Stock}$$
When the question also gives "materials on order", include **both** stock in hand and stock on order on each side.

Cash Budget — the five checks that stop you losing marks

- **Only cash items.** Exclude depreciation, provisions, and any non-cash write-off entirely.
- **Time the receipts.** Prepare a separate **working note for collection from debtors** showing month by month which sale is collected when. Never do this mentally.
- **Time the payments.** "Two months' credit from suppliers" in April means you pay for **February's** purchases. "Wages paid one month in arrears" in April means you pay **March's** wages.
- **Carry the balance forward.** Closing balance of one month becomes the opening balance of the next — including when it is **negative**.
- **A negative closing balance means an overdraft.** Show it in brackets and write the sentence: "the negative balance indicates the extent of overdraft facility required."

Q1 Prepare a budget for 7,000 units from data for 5,000 units

Given (per unit at 5,000 units): Materials 50, Labour 20, Variable overheads 15, Fixed overheads 10 (Rs.50,000 in total), Administrative expenses 10 (5% variable), Selling expenses 6 (20% fixed), Distribution expenses 5 (10% fixed). Total cost of sales per unit Rs.116.

Method — do the splits before you touch the 7,000 column:

1. **Multiply each per-unit figure by 5,000** to get the totals at the present level.
2. **Split the three mixed items** into fixed and variable rupee amounts:
 Administrative: total $10 \times 5,000 = \text{Rs.}50,000 \rightarrow$ variable 5% = Rs.2,500, fixed 95% = Rs.47,500.
 Selling: total $6 \times 5,000 = \text{Rs.}30,000 \rightarrow$ fixed 20% = Rs.6,000, variable 80% = Rs.24,000.
 Distribution: total $5 \times 5,000 = \text{Rs.}25,000 \rightarrow$ fixed 10% = Rs.2,500, variable 90% = Rs.22,500.
3. **Carry every fixed rupee amount across unchanged** to the 7,000 column.
4. **Scale every variable amount by 7,000/5,000 = 1.4** (equivalently, keep the same rate per unit and multiply by 7,000).
5. Show a **Per Unit column in both** sections so the examiner can see the fixed cost per unit falling.

Flexible Budget for the production of 5,000 and 7,000 units

Particulars	Output 5,000 units		Output 7,000 units	
	Per unit	Total (Rs.)	Per unit	Total (Rs.)
Materials	50.00	2,50,000	50.00	3,50,000
Labour	20.00	1,00,000	20.00	1,40,000
Variable overheads	15.00	75,000	15.00	1,05,000
Fixed overheads	10.00	50,000	7.14	50,000

Particulars	Output 5,000 units		Output 7,000 units	
	Per unit	Total (Rs.)	Per unit	Total (Rs.)
Administrative expenses				
Variable (5%)	0.50	2,500	0.50	3,500
Fixed (95%)	9.50	47,500	6.79	47,500
Selling expenses				
Fixed (20%)	1.20	6,000	0.86	6,000
Variable (80%)	4.80	24,000	4.80	33,600
Distribution expenses				
Fixed (10%)	0.50	2,500	0.36	2,500
Variable (90%)	4.50	22,500	4.50	31,500
TOTAL COST OF SALES	116.00	5,80,000	109.94	7,69,600

Answer: Total cost for 7,000 units = **Rs.7,69,600**, i.e. **Rs.109.94 per unit** against Rs.116 at 5,000 units.

Comment to write: the cost per unit falls by Rs.6.06 purely because the same fixed cost of Rs.1,06,000 (50,000 + 47,500 + 6,000 + 2,500) is now spread over 7,000 units instead of 5,000. Variable cost per unit is unchanged at Rs.94.80.

Q2 Flexible budget at 60%, 70% and 90% capacity with stepped semi-variable costs

Given (at 50% capacity): Fixed — Salaries 50,000; Rent and Taxes 40,000; Depreciation 60,000; Administrative expenses 70,000. Variable — Materials 2,00,000; Labour 2,50,000; Others 40,000. Semi-variable — Repairs 1,00,000; Indirect labour 1,50,000; Others 90,000.

Fixed expenses remain constant at all capacities. **Semi-variable expenses do not change between 45% and 61% capacity, rise by 10% between 61% and 75%, and rise a further 5% when capacity crosses 75%.**

Estimated sales: 60% Rs.11,00,000; 70% Rs.13,00,000; 90% Rs.15,00,000.

Method — treat the three cost groups completely differently:

- Fixed group:** copy the same rupee figures into all four columns. No arithmetic at all.
- Variable group:** multiply the 50% figure by (required % ÷ 50). For 60% multiply by 1.2, for 70% by 1.4, for 90% by 1.8.
- Semi-variable group — read the step rule carefully.** It is **not** proportionate:
 - 60% falls inside the 45–61% band → **same as 50%, no change at all.**
 - 70% falls in the 61–75% band → **50% figure + 10%.**
 - 90% is above 75% → **50% figure + 10%, then a further + 5%** (i.e. $\times 1.10 \times 1.05 = \times 1.155$).
- Total the costs, bring in the given sales, and show profit or loss.

Flexible Budget (Rs.)

Particulars	50% capacity	60% capacity	70% capacity	90% capacity
Fixed Expenses				
Salaries	50,000	50,000	50,000	50,000
Rent and Taxes	40,000	40,000	40,000	40,000
Depreciation	60,000	60,000	60,000	60,000
Administrative expenses	70,000	70,000	70,000	70,000
Variable Expenses				
Materials	2,00,000	2,40,000	2,80,000	3,60,000
Labour	2,50,000	3,00,000	3,50,000	4,50,000
Others	40,000	48,000	56,000	72,000
Semi-variable Expenses				
Repairs	1,00,000	1,00,000	1,10,000	1,15,500

Particulars	50% capacity	60% capacity	70% capacity	90% capacity
Indirect Labour	1,50,000	1,50,000	1,65,000	1,73,250
Others	90,000	90,000	99,000	1,03,950
TOTAL COST	10,50,000	11,48,000	12,80,000	14,94,700
Sales	-	11,00,000	13,00,000	15,00,000
Profit / (Loss)	-	(48,000)	20,000	5,300

Answer: a loss of Rs.48,000 at 60% capacity, a profit of Rs.20,000 at 70%, and a profit of Rs.5,300 at 90%.

Comment worth adding: profit is highest at 70% capacity. Beyond 75% the further 5% step in semi-variable costs, together with the smaller price realisation implied by sales of only Rs.15,00,000 at 90%, more than offsets the extra volume. The company should therefore operate around 70% capacity.

Q3 Overhead costs at 50% and 70% capacity, and the overhead recovery rate

Given (at 60% capacity): Variable — Indirect labour 10,500; Indirect materials 8,400. Semi-variable — Repairs and maintenance 7,000 (70% fixed, 30% variable); Electricity 25,200 (50% fixed, 50% variable). Fixed — Office expenses including salaries 70,000; Insurance 4,000; Depreciation 20,000. Estimated direct labour hours at 60% = 1,20,000.

Method:

1. **Split the two semi-variable items at the 60% level first** (working note below). Only the variable half moves.
2. Scale variable amounts by 50/60 and 70/60; keep fixed amounts constant.
3. **Scale the labour hours too** — 1,20,000 at 60% means 1,00,000 at 50% and 1,40,000 at 70%.
4. Overhead recovery rate per hour = Total overhead ÷ labour hours **at that capacity**.

W.N.1 — Segregation of semi-variable overheads at 60% capacity (Rs.)

Item	Total at 60%	Fixed portion	Variable portion
Repairs and maintenance	7,000	4,900 (70%)	2,100 (30%)
Electricity	25,200	12,600 (50%)	12,600 (50%)

Flexible Budget of Overheads (Rs.)

Particulars	50% capacity	60% capacity	70% capacity
Variable Overheads			
Indirect labour	8,750	10,500	12,250
Indirect materials	7,000	8,400	9,800
Semi-variable Overheads			
Repairs and maintenance (4,900 fixed + variable)	6,650	7,000	7,350
Electricity (12,600 fixed + variable)	23,100	25,200	27,300
Fixed Overheads			
Office expenses including salaries	70,000	70,000	70,000
Insurance	4,000	4,000	4,000
Depreciation	20,000	20,000	20,000
TOTAL OVERHEAD COST	1,39,500	1,45,100	1,50,700
Estimated direct labour hours	1,00,000	1,20,000	1,40,000
Overhead recovery rate per hour	1.395	1.209	1.076

Answer: Total overheads — Rs.1,39,500 at 50%, Rs.1,45,100 at 60% and Rs.1,50,700 at 70%. Overhead recovery rates — Re.1.395, Re.1.209 and Re.1.076 per direct labour hour respectively.

Comment: the recovery rate falls as capacity rises because the fixed element of Rs.1,11,500 (70,000 + 4,000 + 20,000 + 4,900 + 12,600) is spread over more hours.

Q4 Budget for production at 80% and 100% capacity

Given (60% activity = 600 units): Materials Rs.100 per unit; Labour Rs.40 per unit; Direct expenses Rs.10 per unit; Factory overheads Rs.40,000 (40% fixed); Administrative expenses Rs.30,000 (60% fixed).

Method:

1. **Convert capacity into units first:** 60% = 600 units, so 80% = **800 units** and 100% = **1,000 units**.
2. Materials, labour and direct expenses are pure variable — just multiply the rate by the units.
3. **Split the two overheads at the 600-unit level:** Factory — fixed 40% = Rs.16,000, variable 60% = Rs.24,000 (Rs.40 per unit). Administrative — fixed 60% = Rs.18,000, variable 40% = Rs.12,000 (Rs.20 per unit).
4. Carry the fixed rupee amounts across unchanged; scale the variable ones by units.
5. Give the per-unit column and watch the fixed cost per unit fall.

Flexible Budget at 60%, 80% and 100% capacity

Particulars	60% — 600 units		80% — 800 units		100% — 1,000 units	
	Total	P.U.	Total	P.U.	Total	P.U.
Materials	60,000	100	80,000	100	1,00,000	100
Labour	24,000	40	32,000	40	40,000	40
Direct expenses	6,000	10	8,000	10	10,000	10
Factory overheads						
Fixed (40%)	16,000	26.67	16,000	20.00	16,000	16
Variable (60%)	24,000	40	32,000	40	40,000	40
Administrative expenses						
Fixed (60%)	18,000	30.00	18,000	22.50	18,000	18
Variable (40%)	12,000	20	16,000	20	20,000	20
TOTAL COST	1,60,000	266.67	2,02,000	252.50	2,44,000	244

Answer: Total cost — **Rs.2,02,000 at 80% (800 units)** and **Rs.2,44,000 at 100% (1,000 units)**; cost per unit falls from Rs.266.67 to Rs.252.50 to Rs.244 as the fixed cost of Rs.34,000 is spread wider.

Q5 Cash budget for April to June 2008

Given: Cash in hand 1.4.2008 Rs.32,000. Monthly figures — Feb: Sales 70,000, Purchases 44,000, Wages 6,000, Expenses 5,000; Mar: 80,000 / 56,000 / 9,000 / 6,000; Apr: 96,000 / 60,000 / 9,000 / 7,000; May: 1,00,000 / 68,000 / 11,000 / 9,000; Jun: 1,20,000 / 62,000 / 14,000 / 9,000.

Suppliers allow **two months'** credit. **25% of sales are for cash**; credit customers take **one month**. Wages and expenses are paid with a **one-month** delay. Income tax Rs.28,000 is payable in June.

Method — build the debtors working note first, then the budget:

1. Split every month's sales: **25% cash in the same month, 75% collected the following month**.
2. Purchases: two months' credit, so April pays for **February**, May pays for **March**, June pays for **April**.
3. Wages and expenses: one month's delay, so April pays **March's** figures, and so on.
4. Opening balance of April is Rs.32,000; each closing balance becomes the next opening balance.

W.N.1 — Collection from sales (Rs.)

Month of sale	Sales	Feb	Mar	Apr	May	Jun
February	70,000	17,500	52,500			
March	80,000		20,000	60,000		
April	96,000			24,000	72,000	
May	1,00,000				25,000	75,000
June	1,20,000					30,000

Month of sale	Sales	Feb	Mar	Apr	May	Jun
Total collection		17,500	72,500	84,000	97,000	1,05,000

Cash Budget for the three months April to June 2008 (Rs.)

Particulars	April	May	June
Opening Cash Balance	32,000	57,000	82,000
Add: Receipts			
Cash sales (25% of the month's sales)	24,000	25,000	30,000
Collection from debtors (75% of previous month)	60,000	72,000	75,000
<i>Total Receipts (A)</i>	<i>1,16,000</i>	<i>1,54,000</i>	<i>1,87,000</i>
Less: Payments			
Payment to suppliers (2 months' credit — Feb / Mar / Apr purchases)	(44,000)	(56,000)	(60,000)
Wages (1 month's delay — Mar / Apr / May)	(9,000)	(9,000)	(11,000)
Expenses (1 month's delay — Mar / Apr / May)	(6,000)	(7,000)	(9,000)
Income tax	-	-	(28,000)
<i>Total Payments (B)</i>	<i>(59,000)</i>	<i>(72,000)</i>	<i>(1,08,000)</i>
Closing Cash Balance (A – B)	57,000	82,000	79,000

Answer: Closing cash balances — April **Rs.57,000**, May **Rs.82,000**, June **Rs.79,000**. The company remains cash-surplus throughout; no overdraft is needed even after paying income tax of Rs.28,000 in June.

Q6 XY Co — cash budget showing the overdraft required each month

Given: Feb — Sales 1,80,000, Purchases 1,24,800, Wages 12,000; Mar — 1,92,000 / 1,44,000 / 14,000; Apr — 1,08,000 / 2,43,000 / 11,000; May — 1,74,000 / 2,46,000 / 10,000; Jun — 1,26,000 / 2,68,000 / 15,000.

50% of credit sales are realised in the month following the sale and the remaining 50% in the second month following. Creditors are paid in the month following purchase. Cash at bank on 1st April Rs.25,000.

Method:

- There are **no cash sales** here — every rupee of sales is collected in two equal halves, one month later and two months later. So April's receipts = 50% of February + 50% of March.
- Creditors: paid the very next month, so April pays for **March's** purchases.
- Wages: no delay is mentioned, so they are paid in the **same** month.
- Carry negative balances forward as they are — that is the whole point of this question.

W.N.1 — Collection from debtors (Rs.)

Month of sale	Sales	Mar	Apr	May	Jun
February	1,80,000	90,000	90,000		
March	1,92,000		96,000	96,000	
April	1,08,000			54,000	54,000
May	1,74,000				87,000
Total collection			1,86,000	1,50,000	1,41,000

Cash Budget for April to June (Rs.)

Particulars	April	May	June
Opening balance	25,000	56,000	(47,000)
Add: Collection from debtors	1,86,000	1,50,000	1,41,000
<i>Total Receipts (A)</i>	<i>2,11,000</i>	<i>2,06,000</i>	<i>94,000</i>

Particulars	April	May	June
Less: Payments			
Payment to suppliers (Mar / Apr / May purchases)	(1,44,000)	(2,43,000)	(2,46,000)
Wages (same month)	(11,000)	(10,000)	(15,000)
Total Payments (B)	(1,55,000)	(2,53,000)	(2,61,000)
Closing balance (A – B)	56,000	(47,000)	(1,67,000)

Answer — extent of bank facility required: April Nil (surplus of Rs.56,000), May an overdraft of **Rs.47,000**, June an overdraft of **Rs.1,67,000**.

Sentence to write: the negative closing balances indicate the extent of the overdraft facility the company must arrange, the maximum requirement being **Rs.1,67,000** at the end of June. The deficit arises because the company is manufacturing for stock — purchases of Rs.2.4–2.7 lakhs a month far exceed the falling sales realisations.

Q7 Cash position at the end of April, May and June 2016

Given: Feb — Sales 1,20,000, Purchases 84,000, Wages 10,000, Misc. 7,000; Mar — 1,30,000 / 1,00,000 / 12,000 / 8,000; Apr — 80,000 / 1,04,000 / 8,000 / 6,000; May — 1,16,000 / 1,06,000 / 10,000 / 12,000; Jun — 88,000 / 80,000 / 8,000 / 6,000.

Sales: **20% realised in the month of sale with a 2% discount**; the balance equally in the two following months. Purchases paid in the month following supply. Wages: **25% paid in arrears** the following month.

Miscellaneous expenses paid a month in arrears. Rent Rs.1,000 per month paid quarterly in advance, due in April. Advance tax Rs.25,000 due by 15th June. Income from investments Rs.5,000 received quarterly in April. Cash in hand 1.4.2016 Rs.5,000.

Method — two working notes are essential here:

- Cash sales carry a 2% discount**, so the cash actually received is 20% of sales × 98%. For April: 80,000 × 20% = 16,000, less 2% = **Rs.15,680**. Do not forget the discount.
- The remaining 80%** is split equally over the next two months, i.e. **40% each**.
- Wages:** 75% in the current month, 25% next month. Show the two lines separately — "wages paid in cash" and "outstanding wages paid".
- Rent:** quarterly in advance in April means **3 × Rs.1,000 = Rs.3,000 in April only**, nothing in May or June.

W.N.1 — Collection from credit sales, i.e. the 80% balance (Rs.)

Month of sale	80% of sales	Apr	May	Jun
February	96,000	48,000		
March	1,04,000	52,000	52,000	
April	64,000		32,000	32,000
May	92,800			46,400
Total collection from debtors		1,00,000	84,000	78,400

W.N.2 — Wages (Rs.)

Month	Total wages	75% paid same month	25% paid next month
March	12,000	9,000	3,000 → April
April	8,000	6,000	2,000 → May
May	10,000	7,500	2,500 → June
June	8,000	6,000	2,000 → July

Cash Budget for April to June 2016 (Rs.)

Particulars	April	May	June
Opening Balance	5,000	5,680	(7,084)
Add: Receipts			

Particulars	April	May	June
Cash sales (20% of sales less 2% discount)	15,680	22,736	17,248
Collection from debtors (W.N.1)	1,00,000	84,000	78,400
Income from investments	5,000	-	-
Total Receipts (A)	1,25,680	1,12,416	88,564
Less: Payments			
Payment to creditors (Mar / Apr / May purchases)	(1,00,000)	(1,04,000)	(1,06,000)
Miscellaneous expenses (one month in arrears)	(8,000)	(6,000)	(12,000)
Wages paid in cash (75% of the month)	(6,000)	(7,500)	(6,000)
Outstanding wages paid (25% of previous month)	(3,000)	(2,000)	(2,500)
Rent (quarterly in advance, 3 × 1,000)	(3,000)	-	-
Advance income tax	-	-	(25,000)
Total Payments (B)	(1,20,000)	(1,19,500)	(1,51,500)
Closing Balance (A – B)	5,680	(7,084)	(62,936)

Answer: Forecast cash position — April a surplus of **Rs.5,680**, May a deficit of **Rs.7,084**, June a deficit of **Rs. 62,936**.

Comment: the position deteriorates sharply in June mainly because of the Rs.25,000 advance tax instalment together with the Rs.12,000 of May's miscellaneous expenses. The company should arrange an overdraft of at least Rs.63,000 by mid-June.

Q8 Month-wise production budget and a summarised production cost budget

Given: No closing WIP is carried. Closing stock of finished goods is held at **50% of the following month's anticipated sales**. Budgeted annual production 20,000 units of Gamma and 25,000 of Delta. Costs per unit — Gamma: Direct material Rs.50, Direct labour Rs.20; Delta: Rs.80 and Rs.30. Other manufacturing expenses apportioned on production — Gamma Rs.2,00,000, Delta Rs.3,75,000.
Estimated sales (units): Gamma — Apr 900, May 1,100, Jun 1,400, Jul 1,800, Aug 2,200, Sep 2,200, Oct 1,800. Delta — 2,900 / 2,900 / 2,500 / 2,100 / 1,700 / 1,700 / 1,900.

Method — the closing-stock rule drives everything:

- Closing stock of any month = 50% of NEXT month's sales.** So April's closing stock = 50% of May's sales. This is why the question gives you October's figures — you need them for September's closing stock.
- Opening stock of any month = closing stock of the previous month.** For April, opening stock = 50% of April's own sales.
- Apply **Production = Sales + Closing Stock – Opening Stock** month by month.
- For the cost budget, first find the **other manufacturing expense rate per unit**: Gamma $2,00,000 \div 20,000 = \text{Rs.}10$; Delta $3,75,000 \div 25,000 = \text{Rs.}15$. Apply to the **half-year production**, not to the annual budget.

(a) Production Budget — GAMMA (units)

Particulars	Apr	May	Jun	Jul	Aug	Sep	Total
Budgeted Sales	900	1,100	1,400	1,800	2,200	2,200	9,600
Add: Desired Closing Stock (50% of next month's sales)	550	700	900	1,100	1,100	900	
Total requirement	1,450	1,800	2,300	2,900	3,300	3,100	
Less: Opening Stock	(450)	(550)	(700)	(900)	(1,100)	(1,100)	
Budgeted Production	1,000	1,250	1,600	2,000	2,200	2,000	10,050

(a) Production Budget — DELTA (units)

Particulars	Apr	May	Jun	Jul	Aug	Sep	Total
Budgeted Sales	2,900	2,900	2,500	2,100	1,700	1,700	13,800
Add: Desired Closing Stock	1,450	1,250	1,050	850	850	950	
<i>Total requirement</i>	<i>4,350</i>	<i>4,150</i>	<i>3,550</i>	<i>2,950</i>	<i>2,550</i>	<i>2,650</i>	
Less: Opening Stock	(1,450)	(1,450)	(1,250)	(1,050)	(850)	(850)	
Budgeted Production	2,900	2,700	2,300	1,900	1,700	1,800	13,300

(b) Summarised Production Cost Budget for the half-year ended 30-09-2002

Particulars	Gamma — 10,050 units		Delta — 13,300 units	
	Cost p.u.	Total (Rs.)	Cost p.u.	Total (Rs.)
Direct Material	50	5,02,500	80	10,64,000
Direct Labour	20	2,01,000	30	3,99,000
Other manufacturing expenses Gamma 2,00,000/20,000 = Rs.10 Delta 3,75,000/25,000 = Rs.15	10	1,00,500	15	1,99,500
Total Production Cost	80	8,04,000	125	16,62,500

Answer: Half-year production — Gamma **10,050 units** costing **Rs.8,04,000**; Delta **13,300 units** costing **Rs. 16,62,500**. Combined production cost **Rs.24,66,500**.

Q9 Quantitative-cum-financial sales budget for the current year

Given (previous year actuals): Zone A — Snow Cream 4,00,000 units @ Rs.12; Talcum Powder 2,50,000 @ Rs. 15; Cold Cream 3,00,000 @ Rs.16. Zone B — Snow Cream 2,50,000 @ Rs.12; Talcum Powder 3,50,000 @ Rs.15; Cold Cream 3,00,000 @ Rs.16.

Current year: Snow Cream up **10% in Zone B**; Cold Cream up **25,000 units in Zone A**; Talcum Powder up **20% in both zones** with a price cut of Re.1. Prices of the two creams are maintained.

Method — handle each product separately and read the wording exactly:

- Snow Cream:** only Zone B changes (+10%): $2,50,000 \times 1.10 = 2,75,000$. Zone A stays at 4,00,000. Price stays Rs.12.
- Talcum Powder:** both zones +20%, and the price falls by Re.1 to **Rs.14**. Zone A $2,50,000 \times 1.20 = 3,00,000$; Zone B $3,50,000 \times 1.20 = 4,20,000$.
- Cold Cream:** Zone A rises by an absolute **25,000 units** (not 25%) to 3,25,000. Zone B unchanged. Price stays Rs.16.
- Show quantity **and** value for every cell — that is what "quantitative-cum-financial" means.

Sales Budget for the current year

Zone	Snow Cream @ Rs.12		Talcum Powder @ Rs.14		Cold Cream @ Rs.16		Total Value (Rs.)
	Units	Value (Rs.)	Units	Value (Rs.)	Units	Value (Rs.)	
Zone A	4,00,000	48,00,000	3,00,000	42,00,000	3,25,000	52,00,000	1,42,00,000
Zone B	2,75,000	33,00,000	4,20,000	58,80,000	3,00,000	48,00,000	1,39,80,000
Total	6,75,000	81,00,000	7,20,000	1,00,80,000	6,25,000	1,00,00,000	2,81,80,000

Answer: Budgeted sales for the current year — **20,20,000 units valued at Rs.2,81,80,000** (Zone A Rs. 1,42,00,000 and Zone B Rs.1,39,80,000).

Note: although the Talcum Powder price is cut by Re.1, the 20% volume rise lifts its value from Rs.90,00,000 to Rs. 1,00,80,000, so the publicity film and price cut together are justified.

Q10 Production budget for the year ended 30 June 2010

Given: Product A — Sales 1,50,000 units, opening stock 14,000, closing stock 15,000. Product B — 1,00,000 / 5,000 / 14,500. Product C — 70,000 / 8,000 / 8,000.

Method: a single formula applied three times — $\text{Production} = \text{Sales} + \text{Closing Stock} - \text{Opening Stock}$. Write the formula at the top of the answer; it is worth a mark by itself.

Production Budget for the year ended 30th June 2010 (units)

Particulars	Product A	Product B	Product C
Budgeted Sales	1,50,000	1,00,000	70,000
Add: Desired Closing Stock (30.06.2010)	15,000	14,500	8,000
<i>Total requirement</i>	<i>1,65,000</i>	<i>1,14,500</i>	<i>78,000</i>
Less: Opening Stock (01.07.2009)	(14,000)	(5,000)	(8,000)
Budgeted Production	1,51,000	1,09,500	70,000

Answer: A **1,51,000 units**, B **1,09,500 units**, C **70,000 units**.

Note that for Product C opening and closing stock are equal, so production exactly equals sales.

Q11 Material procurement budget (quantitative), with materials on order

Given: Estimated sales 40,000 units; each unit needs 3 units of Material A and 4 units of Material B.

Opening balances: Finished products 5,000; Material A 12,000; Material B 20,000. **Materials on order at the beginning:** A 7,000; B 11,000.

Desired closing balances: Finished products 7,000; Material A 15,000; Material B 25,000. **Materials on order at the end:** A 8,000; B 10,000.

Method — three distinct stages, do not merge them:

- Stage 1 — find budgeted production** (in finished units): $\text{Sales} + \text{Closing FG} - \text{Opening FG} = 40,000 + 7,000 - 5,000 = \mathbf{42,000 \text{ units}}$.
- Stage 2 — find materials consumed:** multiply production by the usage per unit. $A = 42,000 \times 3 = 1,26,000$; $B = 42,000 \times 4 = 1,68,000$.
- Stage 3 — find materials to be procured:** $\text{Consumption} + \text{Closing (stock in hand + on order)} - \text{Opening (stock in hand + on order)}$. Materials "on order" count as part of the stock position on both sides — this is the special feature of this problem.

Stage 1: Budgeted Production = $40,000 + 7,000 - 5,000 = \mathbf{42,000 \text{ units}}$

Stage 2: Material A = $42,000 \times 3 = \mathbf{1,26,000 \text{ units}}$; Material B = $42,000 \times 4 = \mathbf{1,68,000 \text{ units}}$

Material Procurement Budget (Quantitative) — units

Particulars	Material A	Material B
Materials consumed in production	1,26,000	1,68,000
Add: Desired closing balance		
Materials in hand	15,000	25,000
Materials on order	8,000	10,000
<i>Total closing balance</i>	<i>23,000</i>	<i>35,000</i>
Less: Opening balance		
Materials in hand	(12,000)	(20,000)
Materials on order	(7,000)	(11,000)
<i>Total opening balance</i>	<i>(19,000)</i>	<i>(31,000)</i>
MATERIALS TO BE PROCURED	1,30,000	1,72,000

Answer: Material A **1,30,000 units** and Material B **1,72,000 units** are to be procured.

Q12 Materials purchase budget (quantitative chart)

Given: Expected sales 50,000 units; each unit needs 2 units of A and 3 units of B. Opening balances — Finished products 10,000; Material A 12,000; Material B 15,000. Desired closing balances — Finished products 14,000; Material A 13,000; Material B 16,000.

Method: identical three-stage drill as Q11, but there are **no materials on order** here, so only stock in hand is adjusted.

Stage 1: Budgeted Production = 50,000 + 14,000 – 10,000 = **54,000 units**

Stage 2: Material A = 54,000 × 2 = **1,08,000 units**; Material B = 54,000 × 3 = **1,62,000 units**

Materials Purchase Budget (units)

Particulars	Material A	Material B
Materials consumed in production	1,08,000	1,62,000
Add: Desired Closing Stock	13,000	16,000
<i>Total requirement</i>	<i>1,21,000</i>	<i>1,78,000</i>
Less: Opening Stock	(12,000)	(15,000)
MATERIALS TO BE PURCHASED	1,09,000	1,63,000

Answer: Material A **1,09,000 units**; Material B **1,63,000 units**.

Q13 Factory overhead cost at 1,40,000 units (segregating semi-variable costs)

Given:

Expense	At 1,20,000 units	At 1,50,000 units
Indirect Materials	2,64,000	3,30,000
Indirect Labour	1,50,000	1,87,500
Maintenance	84,000	1,02,000
Supervision	1,98,000	2,34,000
Engineering Service	94,000	94,000

Required: the cost of each item at 1,40,000 units.

Method — classify each item before computing anything:

- Test for pure variable:** divide the cost by the volume at both levels. If the **rate per unit is the same**, the cost is purely variable. Indirect materials: 2,64,000/1,20,000 = 2.20 and 3,30,000/1,50,000 = 2.20 → **variable**. Indirect labour: 1.25 and 1.25 → **variable**.
- Test for pure fixed:** if the **total is identical** at both levels, it is fixed. Engineering service Rs.94,000 both times → **fixed**.
- Anything left is semi-variable** — Maintenance and Supervision. Segregate them by the high-low method: Variable rate = Change in cost ÷ Change in volume, then Fixed = Total cost – (Variable rate × volume)
- Rebuild each item at 1,40,000 units using its own behaviour.

W.N.1 — Classification and segregation of overheads

Expense	Rate p.u. at 1,20,000	Rate p.u. at 1,50,000	Nature	Variable rate	Fixed portion
Indirect Materials	2.20	2.20	Variable	Rs.2.20	Nil
Indirect Labour	1.25	1.25	Variable	Rs.1.25	Nil
Maintenance	0.70	0.68	Semi-variable	Rs.0.60	Rs.12,000

Expense	Rate p.u. at 1,20,000	Rate p.u. at 1,50,000	Nature	Variable rate	Fixed portion
Supervision	1.65	1.56	Semi-variable	Rs.1.20	Rs.54,000
Engineering Service	—	—	Fixed	Nil	Rs.94,000

Maintenance: variable rate = $(1,02,000 - 84,000) \div (1,50,000 - 1,20,000) = 18,000 \div 30,000 = \text{Re.0.60}$

Fixed = $84,000 - (0.60 \times 1,20,000) = 84,000 - 72,000 = \text{Rs.12,000}$

Supervision: variable rate = $(2,34,000 - 1,98,000) \div 30,000 = 36,000 \div 30,000 = \text{Rs.1.20}$

Fixed = $1,98,000 - (1.20 \times 1,20,000) = 1,98,000 - 1,44,000 = \text{Rs.54,000}$

Factory Overhead Budget at a production of 1,40,000 units

Particulars	Amount (Rs.)
Indirect Materials $(1,40,000 \times 2.20)$	3,08,000
Indirect Labour $(1,40,000 \times 1.25)$	1,75,000
Maintenance	
Fixed	12,000
Variable $(1,40,000 \times 0.60)$	84,000
Supervision	
Fixed	54,000
Variable $(1,40,000 \times 1.20)$	1,68,000
Engineering Service (fixed)	94,000
TOTAL FACTORY OVERHEAD COST	8,95,000

Answer: Total factory overhead at 1,40,000 units = **Rs.8,95,000** (Maintenance Rs.96,000 and Supervision Rs. 2,22,000 in total).

Q14 Master Budget

Given: Sales p.a. — Product A Rs.40,00,000, Product B Rs.60,00,000. Direct material cost = **40% of sales**. Direct wages — 15 workers at Rs.12,000 per month each. Factory overheads — Works manager's salary Rs.20,000 p.m., Foreman's salary Rs.15,000 p.m.; Stores & spares $2\frac{1}{2}\%$ on sales; Depreciation on machinery Rs.1,26,000 p.a.; Light and power Rs.50,000 p.a.; Repairs and maintenance Rs.80,000 p.a.; Miscellaneous expenses **10% of direct wages**; Administration, selling and distribution expenses Rs.1,40,000 p.a.

Method — convert everything to an annual figure first:

- Total sales = $40,00,000 + 60,00,000 = \text{Rs.1,00,00,000}$. **All percentages are applied to this total.**
- Direct material = $40\% \times 1,00,00,000 = \text{Rs.40,00,000}$.
- Direct wages = $15 \text{ workers} \times \text{Rs.12,000} \times \text{12 months} = \text{Rs.21,60,000}$. Multiplying by 12 is the step most often missed.
- Both salaries are monthly — multiply each by 12.
- Stores & spares = $2.5\% \times 1,00,00,000 = \text{Rs.2,50,000}$. Miscellaneous = $10\% \times 21,60,000 = \text{Rs.2,16,000}$ (a percentage of **wages**, not of sales).

Master Budget for the year

Particulars	Rs.	Rs.
Sales		
Product A	40,00,000	
Product B	60,00,000	1,00,00,000
Less: Cost of Sales		
Direct Materials (40% of sales)	40,00,000	
Direct Wages $(15 \times 12,000 \times 12)$	21,60,000	

Particulars	Rs.	Rs.
Works Manager's salary (20,000 × 12)	2,40,000	
Foreman's salary (15,000 × 12)	1,80,000	
Stores and Spares (2½% of 1,00,00,000)	2,50,000	
Depreciation on machinery	1,26,000	
Light and Power	50,000	
Repairs and Maintenance	80,000	
Miscellaneous expenses (10% of direct wages)	2,16,000	
Administration, selling and distribution expenses	1,40,000	(74,42,000)
BUDGETED PROFIT		25,58,000

Answer: Budgeted profit for the year = **Rs.25,58,000** on sales of Rs.1,00,00,000, i.e. a margin of 25.58%.

Q15 Comprehensive set of six budgets for the next fiscal month

Given: Products A, B, C sell at Rs.450, Rs.550 and Rs.650. Sales quantity ratio A : B : C = **1 : 2 : 4**.

Parts per unit — A: D 1, E 10, F 2, G 8 | B: D 1, E 2, F 14, G 10 | C: D 1, E 6, F 10, G 2. Purchase prices: D Rs.45, E Rs.15, F Rs.15, G Rs.5.

Labour hours per unit — A: skilled 6, unskilled 8 | B: 4 and 6 | C: 3 and 6. Wage rates: skilled Rs.6 per hour, unskilled Rs.5 per hour.

Variable overhead per unit produced — A Rs.9, B Rs.11, C Rs.7.

Opening stock — A 500, B 1,000, C 3,000, D 1,500, E 1,000, F 20,000, G 10,000. **Closing stock of products and parts = 90% of opening stock.**

Workers work 8 hours a day for 25 days a month. **Monthly fixed overhead** — A Rs.15,75,000, B Rs.5,80,000, C Rs.8,45,000. **Projected yearly profit Rs.1,20,00,000.**

Method — you cannot start with the sales budget. Work out the sales QUANTITY first:

- Step 1: build the cost card per unit** for each product (raw material + labour + variable overhead) so you can get contribution per unit.
- Step 2: let sales of A be x**, so B = 2x and C = 4x. Express total contribution in terms of x.
- Step 3: form the equation** Total Contribution – Total Fixed Cost = **monthly** profit (= 1,20,00,000 ÷ 12 = Rs.10,00,000) and solve for x.
- Step 4: now prepare the six budgets in sequence** — Sales → Production → Parts usage → Purchase → Manpower → Master. Each one feeds the next, so an error early on cascades.
- Closing stock = 90% of opening stock** for every product and every part — a 10% reduction across the board.

W.N.1 — Raw material (parts) cost per unit

Part	Price (Rs.)	A — nos.	A — cost	B — nos.	B — cost	C — nos.	C — cost
D	45	1	45	1	45	1	45
E	15	10	150	2	30	6	90
F	15	2	30	14	210	10	150
G	5	8	40	10	50	2	10
Raw material cost per unit			265		335		295

W.N.2 — Labour cost per unit

Labour	Rate/hr	A — hrs	A — cost	B — hrs	B — cost	C — hrs	C — cost
Skilled	6	6	36	4	24	3	18
Unskilled	5	8	40	6	30	6	30
Labour cost per unit			76		54		48

W.N.3 — Contribution per unit and computation of sales quantity

Particulars	A	B	C
Selling price per unit	450	550	650
Less: Raw material cost (W.N.1)	(265)	(335)	(295)
Less: Labour cost (W.N.2)	(76)	(54)	(48)
Less: Variable overhead	(9)	(11)	(7)
<i>Total variable cost per unit</i>	<i>(350)</i>	<i>(400)</i>	<i>(350)</i>
<i>Contribution per unit</i>	<i>100</i>	<i>150</i>	<i>300</i>
Sales ratio	x	2x	4x
Total Contribution	100x	300x	1,200x

Total Contribution = $100x + 300x + 1,200x = 1,600x$

Total Fixed Cost = $15,75,000 + 5,80,000 + 8,45,000 = \text{Rs.}30,00,000$ per month

Monthly profit required = $1,20,00,000 \div 12 = \text{Rs.}10,00,000$

$\therefore 1,600x - 30,00,000 = 10,00,000 \Rightarrow 1,600x = 40,00,000 \Rightarrow x = 2,500$ units

Hence sales quantities are **A 2,500, B 5,000, C 10,000 units.**

(a) Sales Budget — quantity and value

Particulars	A	B	C	Total
Sales ratio	1	2	4	
Sales quantity (units)	2,500	5,000	10,000	17,500
Selling price per unit (Rs.)	450	550	650	
Total Budgeted Sales (Rs.)	11,25,000	27,50,000	65,00,000	1,03,75,000

(b) Production Budget (units)

Particulars	A	B	C
Budgeted Sales	2,500	5,000	10,000
Add: Closing Stock (90% of opening)	450	900	2,700
<i>Total requirement</i>	<i>2,950</i>	<i>5,900</i>	<i>12,700</i>
Less: Opening Stock	(500)	(1,000)	(3,000)
Budgeted Production	2,450	4,900	9,700

(c) Parts Usage Budget (nos.)

Product	Production units	Part D		Part E		Part F		Part G	
		No./u	Total	No./u	Total	No./u	Total	No./u	Total
A	2,450	1	2,450	10	24,500	2	4,900	8	19,600
B	4,900	1	4,900	2	9,800	14	68,600	10	49,000
C	9,700	1	9,700	6	58,200	10	97,000	2	19,400
Total parts consumed			17,050		92,500		1,70,500		88,000

(d) Purchase Budget — quantity and value

Particulars	Part D	Part E	Part F	Part G
Parts consumed (c)	17,050	92,500	1,70,500	88,000
Add: Closing Stock (90% of opening)	1,350	900	18,000	9,000
<i>Total requirement</i>	<i>18,400</i>	<i>93,400</i>	<i>1,88,500</i>	<i>97,000</i>
Less: Opening Stock	(1,500)	(1,000)	(20,000)	(10,000)
<i>Parts to be purchased (nos.)</i>	<i>16,900</i>	<i>92,400</i>	<i>1,68,500</i>	<i>87,000</i>
Purchase price per part (Rs.)	45	15	15	5
Purchase Value (Rs.)	7,60,500	13,86,000	25,27,500	4,35,000

Total purchase value = 7,60,500 + 13,86,000 + 25,27,500 + 4,35,000 = **Rs.51,09,000**

(e) Manpower Budget — labour hours and wages payable

Product	Production units	Skilled		Unskilled	
		Hrs/unit	Total hrs	Hrs/unit	Total hrs
A	2,450	6	14,700	8	19,600
B	4,900	4	19,600	6	29,400
C	9,700	3	29,100	6	58,200
Total labour hours			63,400		1,07,200
Wage rate per hour (Rs.)			6		5
Wages payable (Rs.)			3,80,400		5,36,000

Number of workers required (a common add-on part): each worker is available 8 hours × 25 days = **200 hours a month**.

Skilled = 63,400 ÷ 200 = **317 workers**; Unskilled = 1,07,200 ÷ 200 = **536 workers**. Total wages = Rs.9,16,400.

(f) Master Budget for the month

Particulars	Rs.	Rs.
Sales (11,25,000 + 27,50,000 + 65,00,000)		1,03,75,000
Less: Variable Costs		
Parts consumed — D (17,050 × 45)	7,67,250	
Parts consumed — E (92,500 × 15)	13,87,500	
Parts consumed — F (1,70,500 × 15)	25,57,500	
Parts consumed — G (88,000 × 5)	4,40,000	(51,52,250)
Labour — Skilled	3,80,400	
Labour — Unskilled	5,36,000	(9,16,400)
Variable overheads — A (2,450 × 9)	22,050	
Variable overheads — B (4,900 × 11)	53,900	
Variable overheads — C (9,700 × 7)	67,900	(1,43,850)
Add: Cost of goods sold out of Opening Stock (Sales – Production)		
A — 50 units × Rs.350	17,500	
B — 100 units × Rs.400	40,000	
C — 300 units × Rs.350	1,05,000	(1,62,500)
Contribution		40,00,000
Less: Fixed Overheads		
Product A	15,75,000	
Product B	5,80,000	
Product C	8,45,000	(30,00,000)
BUDGETED PROFIT FOR THE MONTH		10,00,000

Answer: the budgeted monthly profit is **Rs.10,00,000**, which annualised gives **Rs.1,20,00,000** — exactly the projected yearly profit. **That agreement is your proof that the whole set of budgets is internally consistent**, and you should state it as the closing line of the answer.

The subtle adjustment in the master budget: production is *less* than sales for every product, so part of the sales comes out of opening stock. Those units carry variable cost but appear nowhere in the current month's material, labour or overhead figures. They must be brought in separately as "cost of goods sold out of opening stock" (A $2,500 - 2,450 = 50$ units; B $5,000 - 4,900 = 100$; C $10,000 - 9,700 = 300$), valued at each product's variable cost of Rs.350, Rs.400 and Rs.350. Miss this and the contribution comes to Rs.41,62,500 and the profit will not tie to Rs.10,00,000.

LAST-MINUTE REVISION SHEET — everything on one spread

Core formulas

Contribution	$\text{Sales} - \text{Variable Cost} = \text{Fixed Cost} + \text{Profit}$
P/V Ratio	$\text{Contribution} \div \text{Sales} \mid \Delta\text{Contribution} \div \Delta\text{Sales} \mid \Delta\text{Profit} \div \Delta\text{Sales} \mid 100\% - \text{VC Ratio}$
BEP	Value: $\text{FC} \div \text{P/V Ratio}$ Units: $\text{FC} \div \text{Contribution p.u.}$
Margin of Safety	$\text{Actual Sales} - \text{BEP Sales} = \text{Profit} \div \text{P/V Ratio}$
Sales for target profit	$(\text{FC} + \text{Desired Profit}) \div \text{P/V Ratio}$ [gross up for tax first: $\text{PBT} = \text{PAT} \div (1 - t)$]
Absorption rate	$\text{Variable production cost p.u.} + (\text{Fixed production OH} \div \text{Normal output})$
Profit reconciliation	$\text{Marginal Profit} + \text{Fixed OH in Closing Stock} - \text{Fixed OH in Opening Stock} = \text{Absorption Profit}$
Limiting factor ranking	Contribution per unit of the limiting factor (per kg / per labour hour / per machine hour)
Shut-down point	$\text{Avoidable Fixed Cost} \div \text{Contribution p.u.}$, where Avoidable FC = FC if operating – (FC if shut + shut-down costs)
Production Budget	$\text{Budgeted Sales} + \text{Desired Closing Stock} - \text{Opening Stock}$
Purchase Budget	$\text{Materials Consumed} + \text{Closing Stock} - \text{Opening Stock}$ (include "on order" on both sides when given)
Semi-variable split	Variable rate = $\Delta\text{Cost} \div \Delta\text{Volume}$; Fixed = $\text{Total Cost} - (\text{Variable rate} \times \text{Volume})$

Decision rules in one line each

Situation	Rule	Sentence to write
Special / export order	Accept if price > variable cost p.u. (including order-specific costs)	"Accept, as it yields a positive contribution of Rs.__ and spare capacity exists."
Same-market order at a low price	Reject regardless of the arithmetic	"Reject, as it would disturb the existing selling price of Rs.__."
Make or buy	Compare buy price + unavoidable fixed cost vs total cost to make	"Buy / Make, as the relevant cost is lower by Rs.__ per unit."
Limiting factor	Rank by contribution per unit of the scarce resource	"__ ranks first, giving Rs.__ contribution per __."
Shut-down	Close if output < shut-down point	"Shut down, as the loss reduces from Rs.__ to Rs.__."
Sunk cost appears	Ignore, but say so	"Book value of Rs.__ is a sunk cost and hence irrelevant."
Permanent labour, redeployed	Ignore, but say so	"Wages are a committed cost with no loss of contribution, hence irrelevant."
Apportioned fixed overhead	Ignore, but say so	"Apportioned overhead involves no additional cash outflow, hence irrelevant."

Corrections applied to the handwritten notes

The handwritten solutions were followed method-for-method, but every figure was re-verified. Seven arithmetical slips were found and corrected. **Use the figures in this document, not the ones in the notebook, for these seven items.**

Where	In the notes	Correct figure	Why
Mod 2, Q8	Year 2 profit shown as 10,000	14,000	36,000 contribution – 22,000 fixed cost = 14,000, which is also the figure given in the question
Mod 2, Q11 case (v)	BEP 13,333; MOS 8,667	BEP 13,200; MOS 8,800	P/V ratio is 54.55%, not 54%. Use $BEP = (FC \div Contribution) \times Sales$ to avoid rounding
Mod 2, Q11 case (vi)	BEP 13,636; MOS 4,364	BEP 13,500; MOS 4,500	P/V ratio is 44.44%, not 44%
Mod 3, Q4	Factory overhead for 3,000 units left at 20,000	30,000	Variable cost must be scaled by $3,000/2,000 = 1.5 \times$
Mod 3, Q6	Conclusion "choose Option 2"	Option 1	The notes' own figures show Option 1 at Rs.14,75,000 against Option 2 at Rs.14,00,000
Mod 3, Q14	Avoidable FC written as $2,00,000 - 80,000 = 1,20,000$	2,00,000 – 88,000 = 1,12,000	FC if shut = 74,000 + 14,000 = 88,000. The notes' own shut-down point of 14,000 tins confirms 1,12,000
Mod 3, Q10	Contribution per labour hour for A shown as 0.95	0.75	$4.50 \div 6 \text{ hours} = 0.75$. The ranking (B first) is unaffected

Coverage of this document

Module	Problems solved	Count
2 — Marginal & Absorption Costing	Q2, Q3, Q4, Q5, Q6, Q7, Q8, Q9, Q10, Q11, Q12, Q13, Q14, Q16, Q17, Q18, Q20	17
3 — Cost-Volume-Profit Analysis	Q1 to Q16 (all)	16
4 — Relevant Costing	Q2, Q3, Q4, Q5, Q6, Q7, Q8	7
5 — Budget & Budgetary Control	Q1 to Q15 (all)	15
Total		55

Not covered, because the handwritten notes do not include them: Module 1 (Process Costing) entirely; Module 6 (Standard Costing & Variance Analysis) entirely; Module 2 Q1, Q15, Q19; Module 3 Q17, Q18; Module 4 Q1 and Q9 to Q15.